

Review article

Understanding the human conflict processing network: A review of the literature on direct neural recordings during performance of a modified stroop task

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ABSTRACT

The Stroop Task is a well-known neuropsychological task developed to investigate conflict processing in the human brain. Our group has utilized direct intracranial neural recordings in various brain regions during performance of a modified color-word Stroop Task to gain a mechanistic understanding of non-emotional human conflict processing. The purpose of this review article is to: 1) synthesize our own studies into a model of human conflict processing, 2) review the current literature on the Stroop Task and other conflict tasks to put our research in context, and 3) describe how these studies define a network in conflict processing. The figures presented are reprinted from our prior publications and key publications referenced in the manuscript. We summarize all studies to date that employ invasive intracranial recordings in humans during performance of conflict-inducing tasks. For our own studies, we analyzed local field potentials (LFPs) from patients with implanted stereotactic electroencephalography (SEEG) electrodes, and we observed intracortical oscillation patterns as well as inter-cortical temporal relationships in the hippocampus, amygdala, and orbitofrontal cortex (OFC) during the cue-processing phase of a modified Stroop Task. Our findings suggest that non-emotional human conflict processing involves modulation across multiple frequency bands within and between brain structures.

1. Introduction

The Stroop Task is a well-studied color-word conflict neuropsychological paradigm that assesses conflict processing, defined as when the processing of one attribute of a stimulus affects the simultaneous processing of another quality of the same stimulus (Scarpina and Tagini, 2017). In its classic manifestation, proposed by John Ridley Stroop in 1935, two types of stimuli were presented to subjects: a congruent condition (e.g., the word “blue” printed in blue ink) and an incongruent condition (e.g., the word “blue” printed in red ink) (Fig. 1). The subject must perform a cognitively demanding task (i.e., naming the ink color) while inhibiting interference from a less cognitively demanding task (i.e., reading the name of the color). The difficulty in inhibiting the more

automated process is called the Stroop effect (Stroop, 1935).

The Stroop task was originally designed to assess cognitive interference skills in patients with ventromedial prefrontal cortex damage (Stroop, 1935). Other conflict processing tasks have also been developed, including (i) the flanker task, in which the orientation of flanking arrows distracts from judgment of a central target arrow (Sanders et al., 2018); (ii) spatial conflict or Simon tasks, in which stimulus response incompatibility is generated by reacting to visual stimuli with an opposing physical response (e.g., requiring a right-handed response to a left-sided stimulus) (Lu and Proctor, 1995; Simon and Berbaum, 1990; Simon, 1990), and (iii) the multi source interference task (MSIT), which is a commonly used computerized task that includes a combination of flanker and Simon tasks (Bush and Shin, 2006). However, conflicts

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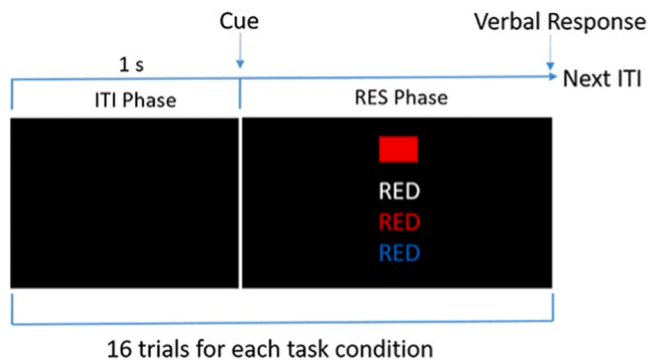


Fig. 1. Conditions within our modified Stroop Task. The four conditions of our group's modified Stroop Task are the following (from the top down) with correct responses underlined and in parentheses: (1) name the color of the rectangle (15 cm×10 cm) as it appears (RED); (2) read the text presented in the white font (RED); (3) respond with the font color when the text is congruent (RED); (4) respond with the font color when the text is incongruent (BLUE). Figure adapted from [Chen et al. \(2020\)](#). ([Chen et al. 2020](#)).

ensuing from each task are resolved by different mechanisms. For example, a study by [Scerrati et al. \(2017\)](#) showed that the Stroop conflict and Simon conflict processing occur by independent pathways ([Scerrati et al., 2017](#)). We have chosen to utilize the Stroop task in our studies because it is a well-known behavioral task widely used to study the brain and characterize neural and behavioral reflections of conflict resolution ([Oehrn et al., 2014](#); [Chen et al., 2020](#); [Fu et al., 2019](#); [Tang et al., 2016](#); [Bartoli et al., 2018](#); [Fu et al., 2022b](#); [Xiao et al., 2023](#)). Other iterations of the Stroop task have been developed, including an auditory Stroop task ([Donohue et al., 2012](#)) and a counting Stroop task ([West and Bailey, 2012](#)). However, the original color-word Stroop task seems consistently more sensitive to post-conflict cognitive control recruitment than its auditory and counting adaptations ([Larson, Clayton and Clawson, 2014](#)).

Since its development, the role of the Stroop task has been expanded and applied to study many aspects of cognitive function. Previous literature has shown that non-emotional conflict response requires multiple parallel processes, including conflict detection, resolution, and adaptation ([Tanaka, Buckley and Mansouri, 2009](#); [Botvinick et al., 2001](#)). Therefore, the Stroop task has also been used as an executive function neuropsychological test both in patients with Parkinson's disease before and after Deep Brain Stimulation (DBS) surgery ([Jahanshahi, Czernecki and Zurowski, 2011](#)) and in patients with neuropsychiatric disorders such as Obsessive Compulsive Disorder (OCD) ([Doolub et al., 2023](#)) and Alzheimer's disease ([Goldberg and Bougakov, 2005](#); [Perry, Watson and Hodges, 2000](#)).

Invasive or noninvasive methods are required to observe neural activity during performance of the Stroop task. Functional magnetic resonance imaging (fMRI) is frequently employed to visualize cortical activation during motor, sensory, memory, recognition, and other pertinent brain functions ([Logothetis, 2008](#)). fMRI studies utilizing the Stroop task have revealed that conflict mainly activates frontal brain areas like the anterior cingulate cortex (ACC) and the dorsolateral prefrontal cortex (DLPFC) ([Hanslmayr et al., 2008](#)). fMRI allows for good visualization of large brain networks, but it suffers from limitations in temporal resolution. On the other hand, when clinically indicated, direct neural recordings have the benefits of increased temporal resolution compared to fMRI and precise visualization of brain signaling during task performance ([Sato et al., 2012](#); [Jung et al., 2006](#)). Other noninvasive paradigms including positron emission tomography (PET) ([Mintun et al., 1995](#)) and magnetoencephalography (MEG) ([Galer et al., 2014](#)) have all been used to identify and record neuronal substrates of selective attention or conflict during the Stroop task. Additionally, EEG recordings have been used in the context of the Stroop task to discern

whether the posterior or anterior attention systems are more sensitive to age-related decline ([West and Bell, 1997](#)). Although these modalities have the benefit of being non-invasive, they do not afford the same temporal and spatial resolution of direct neural recordings. Electro-corticography (ECoG), which does offer recordings from a large cortical surface area and allows for very good spatial resolution, has been used to characterize the Stroop response in epileptic patients ([Koga et al., 2011](#)), but these kinds of studies are limited in their ability to record from deep structures. Stereotactic electroencephalography (SEEG), an invasive surgical technique traditionally used to monitor focal electrophysiologic brain signals in patients with epileptic seizures, enables monitoring of neural activity in deeper neural structures and permits subsequent analyses ([Reif, Strzelczyk and Rosenow, 2016](#)). Ultimately, when a part of the clinical procedure, placing electrodes in carefully selected and targeted brain areas of patients with medically refractory epilepsy allows for detecting specific electrophysiologic signals important for seizure localization and provides an enhanced understanding of the neural substrates of human behavior in patients who wish to participate in behavioral studies.

The two main systems necessary to evaluate goal-directed behavior are regulative and evaluative processes ([Botvinick et al., 2001](#); [Botvinick, Cohen and Carter, 2004](#); [Kerns et al., 2004](#); [MacDonald Iii et al., 2000](#); [Perlstein et al., 2006](#)). Regulative control processes exercise top-down control on sensory processing to complete and adjust to task demands, allocate attention, and override task-inappropriate responses, while evaluative processes include monitoring performance for conflict and providing feedback about control mechanisms ([Larson, Clayton and Clawson, 2014](#); [Egner and Hirsch, 2005](#)). Stroop task neural oscillations have implicated the dorsolateral prefrontal cortex (DLPFC) and ventrolateral prefrontal cortex (VLPFC) in regulative control processes, while evaluative control processes seem to originate in the anterior cingulate cortex (ACC) ([Egner, 2011](#); [Egner and Hirsch, 2005](#); [Kerns et al., 2004](#); [MacDonald Iii et al., 2000](#); [Ridderinkhof et al., 2004](#)). Numerous groups have recorded neural oscillations from the structures of the medial prefrontal cortex [MFC, including the ventromedial prefrontal cortex (VMPFC), pre-supplementary motor area (pre-SMA), SMA, and dorsal ACC (dACC)] ([Koga et al., 2011](#); [Sheth et al., 2012](#); [Bonini et al., 2014](#); [Caruana et al., 2014](#); [Oehrn et al., 2014](#); [Tang et al., 2016](#); [Bartoli et al., 2018](#); [Smith et al., 2019](#); [Ebitz et al., 2020](#); [Fu et al., 2019](#); [Xiao et al., 2023](#)). Our group has built upon these studies and used SEEG to record oscillation patterns during non-emotional conflict processing Stroop tasks, through which we provided novel information on other structures such as the OFC ([Tang et al., 2021a](#)), hippocampus ([Chen et al., 2020](#)), and amygdala ([Tang et al., 2021b](#)). We compare our results with other studies of human conflict processing in the same brain structures in addition to the subthalamic nucleus (STN) and additional cortical regions. Specifically, our studies examine oscillation patterns in the theta (4–8 Hz), beta (13–30 Hz), and gamma (30–200 Hz) frequency ranges, as well as intercortical temporal relationships, such as phase amplitude coupling (PAC), during the cue-processing phase of the Stroop task. Elucidating the relationships between electrophysiological signaling and information processing will improve the understanding of coherence in human conflict resolution.

2. Cortical regions

Neural oscillations during the Stroop task have been investigated extensively in the ACC ([Sheth et al., 2012](#); [Fu et al., 2019](#); [Tang et al., 2016](#); [Bartoli et al., 2018](#); [Smith et al., 2019](#); [Ebitz et al., 2020](#); [Fu et al., 2022b](#)) and multiple regions of the PFC, including the DLPFC ([Oehrn et al., 2014](#)), dorsomedial prefrontal cortex (DMPFC) ([Oehrn et al., 2014](#)), and OFC ([Tang et al., 2021a](#)). Early evidence concerning the roles of the PFC in human conflict processing was demonstrated by decreased Stroop task performance in the setting of various cortical lesions ([Vendrell et al., 1995](#); [Turken and Swick, 2008](#)). In the 1990 s, functional neuroimaging studies also identified the ACC as an important structure

in executive cognitive control (Posner and Dehaene, 1994; Carter et al., 1998). Based in literature focusing on brain activation techniques including functional neuroimaging and event-related potentials, Botvinick et al. (2001) used computational models to propose the conflict monitoring hypothesis, which states that the ACC monitors information and, upon detecting a conflict, alerts other brain regions to resolve this conflict (Botvinick et al., 2001; van Veen and Carter, 2002; Botvinick, Cohen and Carter, 2004). The study also speculated that the PFC might be involved in cognitive control due to its strong anatomical connections to the ACC and the co-occurrence of PFC and ACC activation in previous functional neuroimaging studies (Botvinick et al., 2001). Kerns et al. (2004) corroborated several aspects of the conflict monitoring hypothesis by recording fMRI activity in the ACC and PFC during the Stroop task (Kerns et al., 2004). The authors found that greater ACC activity during high conflict and error trials was associated with behavioral adjustments in the following trial. Additionally, trials following greater ACC activity displayed increased PFC activity responsible for the execution of cognitive control.

In 2007, Carter and van Veen expanded upon the conflict monitoring hypothesis with the conflict-control loop theory (Carter and van Veen, 2007), which states that the ACC detects conflict and specifically recruits the DLPFC for conflict resolution. They supported their theory with evidence from both event-related potential (ERP) and fMRI studies. Their theory was partly based on previous studies of the Stroop task, such as MacDonald et al. (2000), which used event-related fMRI to demonstrate that the ACC and DLPFC function independently during a modified Stroop Task (MacDonald Iii et al., 2000). The results of this study demonstrated greater left DLPFC activity during color naming than with word reading, reflecting its role in the execution of cognitive control. Conversely, the ACC had greater activity during incongruent trials than congruent trials, reflecting its role in conflict monitoring. Therefore, MacDonald et al. (2000) concluded that the ACC and DLPFC have distinct but complementary roles as part of a neural network in conflict processing. Specifically, conflict processing is believed to be carried out by the ACC, while the allocation of attentional resources necessary to resolve conflict and performance adjustment is thought to depend on the PFC (Gruber and Goschke, 2004; Kerns et al., 2004; Casey et al., 2000; MacDonald Iii et al., 2000; Bunge et al., 2002; Rowe et al., 2000; Botvinick et al., 2001; van Veen and Carter, 2002; Haupt et al., 2009; Botvinick, Cohen and Carter, 2004; Carter and van Veen, 2007). Several fMRI studies have further supported the role of the ACC in detecting and processing cognitive conflict and that of the DLPFC in inhibition control and conflict resolution (van Veen and Carter, 2005; Carter et al., 2000; Botvinick, 2007).

Studies of the Stroop task have also used fMRI data to analyze intraregional activity of brain structures involved in conflict processing, such as the ACC and PFC. Haupt et al. (2009) used an auditory version of the Stroop task (i.e., conflict interference between tone pitch and word meaning) to examine the caudal and posterior aspects of the ACC (Haupt et al., 2009). Their results showed that these two parts of the ACC were only activated during task-related interference in the auditory Stroop task (Haupt et al., 2009). Regarding the PFC, Van Veen and Carter et al. (2005) indicated that separate areas of the PFC can resolve semantic or intermediate-level conflict and response conflict by utilizing different semantic conflict and response conflict versions of the Stroop task (van Veen and Carter, 2005). Using fMRI recordings of their participants ($n=14$), the authors found that the superior DLPFC was more activated by semantic conflict, while the inferior DLPFC was more activated by response conflict.

To understand the temporal relationships in conflict processing, several EEG and SEEG studies began to examine intercortical neurodynamics involved in the Stroop task. In a study by Hanslmayr et al. (2008), localization analysis of scalp EEG recordings of their participants ($n=21$) showed that the magnitude of theta (4–7 Hz) oscillations in the ACC is positively correlated with the level of cognitive interference (Hanslmayr et al., 2008). In this study, the authors defined a higher

interference level as synonymous with incongruent trials and a lower interference level as synonymous with congruent and neutral trials. Additionally, they found that theta phase coupling between the ACC and left PFC persisted longer in the incongruent condition of the Stroop task, which indicates that the interaction between these two areas is involved in conflict processing. To investigate the neural markers of conflict processing, Heidlmayr et al. (2020) provided a review of EEG markers related to different cognitive stages during Stroop executive control processes (Heidlmayr, Kihlstedt and Isel, 2020). Consistent with previous fMRI studies, a fronto-central N2 component was found to reflect conflict monitoring processes, with its main neural generator being the ACC. Then, following this N2 mark, a centro-posterior N400 and a late sustained potential (LSP) produced by the PFC is thought to represent interference suppression, whereas the LSP plausibly reflects conflict resolution processes.

To expand upon these findings and gain more insights into the single-neuron or multi-unit involvement of cortical structures in conflict processing and error monitoring, numerous studies have used invasive direct neural recordings from cortical structures during performance of the Stroop, MSIT, Simon, and flanker tasks. We present a detailed summary of some of the most important studies reporting intracranial recordings from cortical structures during these conflict-processing tasks. One key feature shared between these studies is the observation that reaction times are longer in the incongruent trials than in congruent trials, so behavioral effects, although universally reported in these studies, will not be discussed except for as they relate to neural recordings.

The first known study to use intracranial recordings in human subjects during performance of a Stroop task was by Koga et al. (2011) (Koga et al., 2011). Using recordings from subdural electrodes, they evaluated gamma-band modulation in the dorsolateral premotor cortex, DLPFC, and SMA in the color-word Stroop task in 5 pediatric patients undergoing seizure localization surgery. 7 sites in the left DLPFC, 2 sites in the left dorsolateral premotor cortex, and one site in the right medial superior frontal area showed gamma modulation ($p < 0.05$, band-width > 20 Hz, > 20 ms duration, cluster permutation test) during the incongruent condition, occurring 500–200 ms before response. Interestingly, they also performed functional stimulation mapping by stimulating the sites that showed significant gamma modulation (50 Hz, 300 μ s pulse width, 3–9 mA, 5 second maximum duration). They found that upon stimulation of 4 of 9 areas (44%, all in the left dorsolateral premotor area), participants showed temporary naming deficits, which was much greater than when non-Stroop modulating areas were stimulated ($p = 0.007$, Fisher's exact test). These results indicate the role of gamma-band modulation in the dorsolateral premotor cortex, DLPFC, and SMA in conflict processing.

Sheth and colleagues (2012) investigated the neuronal circuitry in the MSIT at the single neuron level (Sheth et al., 2012). Their study combined preoperative fMRI and intraoperative single-cell direct neural recordings from the dACC that were obtained using microelectrode recordings during stereotactic cingulotomy for treatment-refractory OCD (59 individual neurons, $n = 6$ patients). In the fMRI analysis, they found higher BOLD signal in the dACC and DLPFC during trials that involved interference. From the single neuron recordings, they found that there were three distinct cell populations that varied in their firing rates with respect to timing in the task: either before cue presentation (20%), after cue presentation (41%), or after behavioral choice (39%). The cue-responsive neurons showed higher firing rates with increasing levels of interference ($p = 6.4 \times 10^{-3}$, Mann-Whitney test). They also found that there was a “history-dependence” such that noninterference trials preceded by noninterference trials had shorter reaction times than when preceded by interference trials, and interference trials preceded by interference trials had shorter reaction times than when preceded by noninterference trials. These results are consistent with the “Gratton effect,” which states that conflict effects are larger in congruent relative to incongruent trials (Gratton, Coles and Donchin, 1992). Interestingly,

after cingulotomy, the history-dependent changes in reaction times were abrogated (Fig. 2). These results indicate that neurons in the dACC not only detect conflict but also integrate information from the recent past in order to efficiently allocate cognitive resources so that efficient action can be optimized.

In addition to studying successful conflict processing, investigating unsuccessful conflict processing or self-correction of errors can provide valuable information. Bonini et al. (2014) investigated LFPs in the frontal lobe during error trials in a Simon task using SEEG in 5 participants undergoing seizure localization (Bonini et al., 2014). They performed a response-locked analysis that made use of electromyogram (EMG) in the hands, which helped differentiate between overt errors (incorrect button push with full amplitude) and covert errors (incorrect button push with partial amplitude followed by self-corrected response). Error-evoked LFPs were observed in the medial frontal lobe and peaked between 100 and 190 ms after EMG firing. Overt errors produced the largest LFPs (95 μ V), followed by covert errors (68 μ V) and correct trials (26.8 μ V). The strongest modulation was observed in the posterior mesial frontal lobe within the SMA, and surprisingly, electrodes in the rostral cingulate zone did not show modulation. There was also some modulation in the pregenual ACC and the orbitomedial prefrontal cortex (OMPFC), but this activity was delayed compared to the SMA, in which activity had a caudo-rostral latency gradient and was more spatially diffuse. Latencies from single-trial LFPs from the SMA and mesial prefrontal regions were positively correlated ($\rho = 0.8$, $P < 0.01$ between SMA and posterior ACC in one patient, and $\rho = 0.35$, $P < 0.05$ between SMA and OMPFC in another patient, linear regression), and amplitudes were also positively correlated, though less strongly ($\rho = 0.63$, $P < 0.01$ and $\rho = 0.28$, $P < 0.1$, for patient 3 and 5, respectively, linear regression). The SMA LFP was always present in the error response, but activity in other medial prefrontal areas was sometimes absent, suggesting a hierarchy between the regions. In covert errors, the LFP began after EMG firing from the partial incorrect response and continued until the initiation of the corrected response. Taken together, the results indicate that the SMA plays a key role in error monitoring.

Building off the work by Bonini et al. (2014) and Fu et al. (2019) investigated error response at the single neuron level in the dACC and the pre-SMA in 29 patients (399 single units in dACC, 431 in pre-SMA) with SEEG electrodes implanted as a part of seizure localization (Fu et al., 2019). The subjects performed a color-naming Stroop task, and single unit spikes were recorded. As expected, the authors found increased reaction times in incongruent tasks and longer reaction times in correct trials following error trials. 34% ($n = 134$) of dACC neurons and 46% ($n = 198$) of pre-SMA neurons were active in response to errors, and they observed the maximal spike rate after the incorrect action but

before the external feedback. Of these error responsive neurons, 99 neurons (74%) in the dACC and 118 neurons (60%) in the pre-SMA had higher spike rates during error than during correct responses (“type I error neurons”), and 35 neurons (26%) in the dACC and 80 neurons (40%) in the pre-SMA had lower spike rates during error trials than for correct trials (“type II error neurons,” $p < 10^{-10}$, t-test). Spike rates in error neurons did not distinguish between incongruent and congruent conditions. A second group of neurons (“conflict neurons”) had increased spike rates after stimulus onset in incongruent compared to congruent trials [$n = 43$ (7%) of all neurons in the dACC and $n = 54$ (10%) of all neurons in the pre-SMA]. Most error neurons were not conflict neurons. A third group of neurons, the “error-integrating neurons,” whose spike rate signaled whether the previous trial was an error or not, comprised 11.5% ($n = 46$) of neurons in the dACC and 13.5% ($n = 58$) of neurons in the pre-SMA. These neurons had a sustained peri-stimulus error signal that was reinforced at the onset of the subsequent trial. Some of these error-integrating neurons were also error neurons (12 in the dACC and 20 in the pre-SMA). On a group level, the authors compared spike rates between the correct incongruent and the error incongruent conditions and found that only type II error neurons in the dACC ($p = 0.006$, rank-sum) and conflict neurons in the dACC ($p < 0.001$, rank-sum) and pre-SMA ($p = 0.034$, rank-sum) had differential firing between conditions. Using waveform analysis, they found that most relevant neurons (80%) had spike characteristics consistent with pyramidal cells, and most were excitatory. They determined that error signal latencies in the pre-SMA (median 110 ms) occurred significantly earlier ($p = 0.002$, rank sum) than in the dACC (median 165 ms). The authors also demonstrated that single unit intracranial correlated negativity (iERN), a negative deflection signal after response, correlated with the scalp EEG ERN and was present in the dACC and pre-SMA. In addition, the spike rate was different between error and correct trials ($p < 10^{-10}$, signed rank). Finally, the authors conducted a time-frequency analysis and found that “slow theta (2–5 Hz)” and “theta (5–10 Hz)” power increased in error and correct trials, with a stronger increase in error trials in the dACC ($p < 10^{-5}$) and pre-SMA ($p < 10^{-5}$, signed rank), and that slow-theta and theta power decreased in the hippocampus ($p = 0.01$, signed rank). However, there was no difference between congruent and incongruent error trials. The theta power increases correlated with iERN peak amplitude ($p < 10^{-10}$, t-test), and the iERN occurred earlier in the pre-SMA than the dACC by 40 ms ($p < 10^{-10}$, rank sum). Using a multilevel model, they found that spike rates after action onset in type I error neurons, but not type II, correlated with iERN amplitude ($p = 0.001$ in dACC and $p < 0.001$ in pre-SMA, permutation test). Error neuron spike rates were not correlated with reaction time, but the iERN amplitude was associated with shorter reaction times in the dACC and pre-SMA ($p = 0.0001$ for dACC and $p = 0.031$ for pre-SMA, likelihood ratio test). Using a mixed effect model, the authors found that a higher correlation between iERN and spike rate had a greater correlation with post-error slowing ($p = 0.015$, cluster permutation). Error-integrating neuron spike rate in a trial following an error trial also had strong correlation with post-error slowing in the dACC ($p < 0.001$, cluster-based permutation test) but not in the pre-SMA. Overall, the work by Fu et al. (2019) offered considerable insights into the role of the dACC and pre-SMA in error processing at the single neuron and multi-unit levels.

To elucidate neural activity during the flanker test, Caruana et al. (2014) recorded high gamma oscillations (50–150 Hz) in the PMC during performance of a bimanual version of the Ericksen flanker test from SEEG contacts in 6 patients undergoing seizure localization (Caruana et al., 2014). They found that 26 of 46 contacts showed a significant response ($p < 0.05$, t-test) in the incongruent compared to the congruent condition [14 in the dorsal PMC (dPMC), 10 in Brodmann area 6 (mBA6, SMA), and 2 in the ventral PMC (vPMC)]. In a repeated measures ANOVA, they found that there was a significant effect of time ($p < 0.001$, Bonferroni corrected) and condition*time ($p < 0.05$, in 7/26 contacts). Post-hoc analysis showed a significant increase in gamma power in the incongruent vs. congruent and in neutral conditions. They

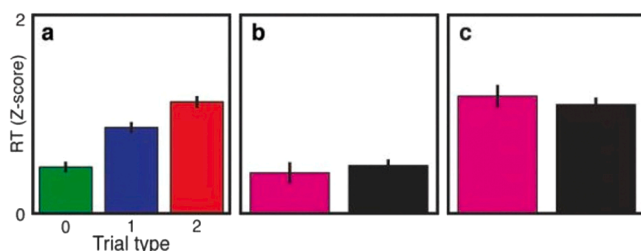


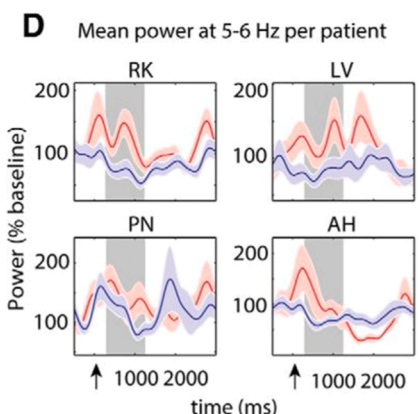
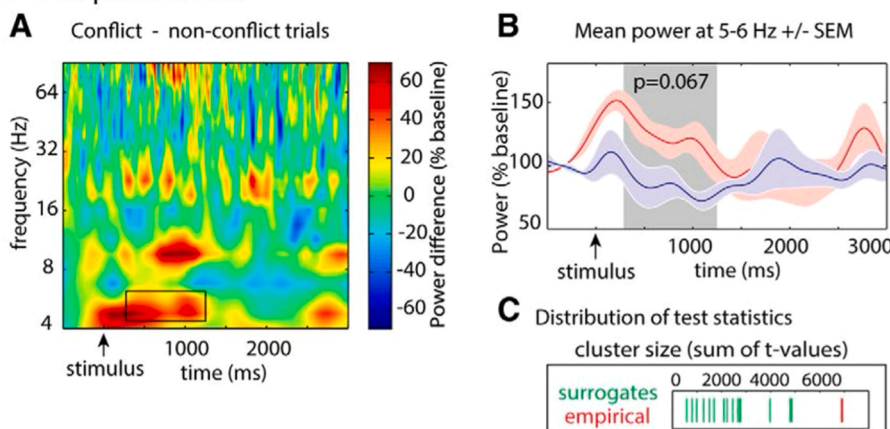
Fig. 2. Abolition of behavioral adaptation following a targeted dACC lesion. RTs were recorded following cingulotomy, in which a stereotactic lesion was created precisely in the region of the dACC from which fMRI signals and microelectrode recordings were obtained. (A) RTs followed a dose-response pattern by trial type ($p < 1 \times 10^{-12}$, ANOVA) similar to that before the lesion (Fig. 1C). Error bars represent s.e.m. ($n = 572$). Behavioral adaptations (the influences of previous trial identity on current trial reaction times), however, were abolished for both (B) non-interference ($p = 0.54$) and (C) high-interference ($p = 0.53$) trials. Figure reproduced from Sheth et al. (2012). (Sheth et al. 2012).

also evaluated the effects of gamma-band modulation from ipsilateral vs. contralateral brain areas and found a significant interaction between condition*time ($p < 0.05$, ANOVA with Bonferroni correction) in 17 of 46 contacts (8 dPMC, 2 vPMC, 7 mBA6). Post-hoc analysis showed longer activity during contralateral trials for the dPMC and vPMC, suggesting roles in movement execution. mBA6 electrodes also showed stronger activity in trials with the ipsilateral hand, suggesting a role for mBA6 in the inhibition of unrequested movement. This work established the importance of gamma-band activity in the PMC in conflict processing.

In a study by Oehrns et al. (2014), a combination of SEEG depth electrodes and subdural strip/grid electrodes were used to explore the oscillatory patterns within and between the DMPFC and DLPFC during an auditory Stroop task in 12 patients with epilepsy (Oehrns et al., 2014). The authors found that successful conflict processing required DMPFC theta (4–8 Hz) and DLPFC gamma (30–100 Hz) modulation (Fig. 3). DMPFC theta power was detected early (between 290 and 1251 ms) and had a positive correlation with reaction time ($t_{(3)} = 3.9$, $p < 0.05$ for incongruent; $t_{(3)} = 1.0$, $p = 0.38$ for congruent, permutation test), suggesting its main role is conflict detection. In the DLPFC, conflict was

associated with increases in gamma power ($F = 6.6$, $p < 0.05$, ANOVA), but there was no conflict modulation in the theta range. They evaluated phase-synchronization between the DMPFC and DLPFC ($n = 4$ patients) in theta (5–6 Hz), alpha (9–12 Hz), and beta (13–29 Hz) frequencies, and they found that conflict was associated with increased synchronization between the DMPFC and the DLPFC at 5 Hz both before and after the average response latency ($p = 0.067$, the minimum value achievable in the test, cluster permutation test). In a comparison between correct and incorrect conflict trials with respect to pre-response phase-synchronization between the DMPFC and DLPFC at 5 Hz, the authors found that there was a significant result ($p = 0.067$, cluster permutation), indicating that DMPFC-DLPFC communication during conflict trials predicts correct conflict resolution. In addition, the cross-frequency coupling (CFC) findings revealed increased phase-amplitude coupling (PAC) between theta phase and gamma amplitude within the DLPFC during an early time window (10–300 ms after the stimulus) in successful incongruent trials ($p < 0.001$, cluster permutation test). This intra-regional PAC found in the DLPFC is hypothesized to reflect successful conflict processing, based on previous studies suggesting that DLPFC activity influences subsequent accuracy or reaction times in

DMPFC power effects



DLPFC power effects

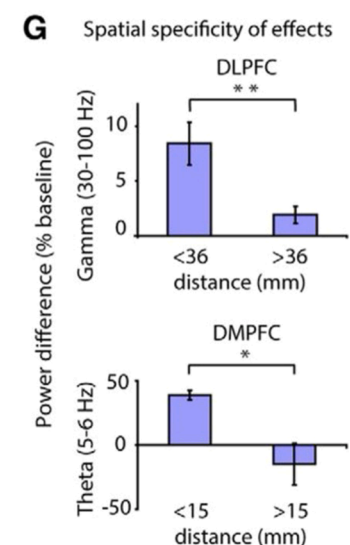
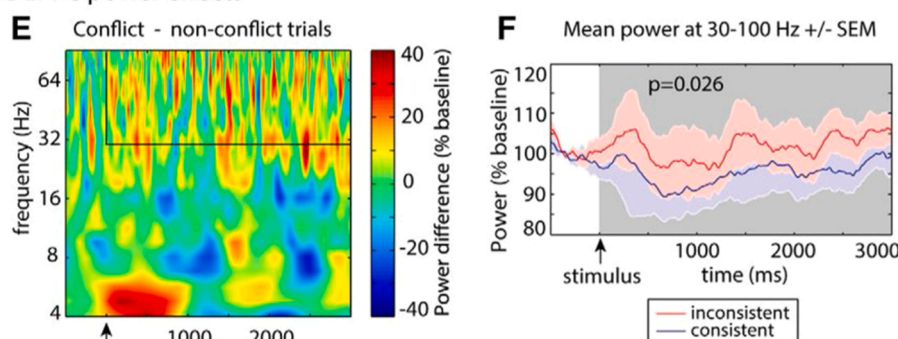


Fig. 3. DMPFC theta enhancement of DLPFC gamma is required for successful conflict processing a phonetic Stroop task designed by Oehrns et al. (2014). (A,E) Both display time-frequency plots during conflict processing, with significant clusters highlighted. (B,F) Both display time series of mean power $\pm$ standard error of the mean across theta and gamma frequencies for conflict and nonconflict stimuli. (C) An illustration of the location of the observed DMPFC theta power test statistic. (D) These plots display conflict-related mean theta-power in the DMPFC for all patients with electrodes in this region, with significant time periods shaded in gray. (G) This bar plot displays conflict-related power increases in near vs distant electrodes from a respective target coordinate across all patients. This figure was reproduced from Oehrns et al. (2014). (Oehrns et al. 2014).

Stroop task paradigms (MacDonald *et al.*, 2000; Stern and Mangels, 2006; Coste *et al.*, 2011). They also found that pre-response CFC in the DLPFC was predictive of a correct vs. incorrect trial from 361 to 464 ms at 4–5 Hz ($p < 0.05$, cluster permutation test). There was also inter-regional CFC between DLPFC gamma power and DMPFC theta phase for conflict vs. non-conflict trials before and after mean reaction time ($p = 0.067$, cluster permutation test), and this CFC was statistically significant when comparing correct and incorrect conflict trials at 5 Hz between 603 and 630 ms ($p = 0.067$, cluster permutation test). Using Granger Causality analysis for correct trials only, they found that there was information transfer in the theta range from the DMPFC to the DLPFC earlier in the post-stimulus period (60–616 ms), and then from the DLPFC to the DMPFC in later time points (808–1408 ms). Altogether, the results from Oehr *et al.* (2014) results expand on the conflict-control loop theory (Carter and van Veen, 2007) by reinforcing the DLPFC's role in conflict resolution and identifying the DMPFC as an additional structure for conflict detection.

Tang and colleagues (2016) recorded LFPs from SEEG contacts during performance of a color-word Stroop task (participants state the color of the word) and a Reading task (participants read a word that is presented in different colors) in 15 patients with epilepsy (Tang *et al.*, 2016). They focused their analysis on low gamma frequency modulation (70–120 Hz) in the medial frontal cortex (mFC), DLPFC, OFC, and ACC in a response-locked analysis. They found that gamma-band power was significantly higher in incongruent trials ($p < 10^{-5}$, ANOVA), which was specific for the Stroop task and not the Reading task. The maximal gamma power was also correlated with the behavioral reaction time for incongruent trials ($\rho = 0.25$ Pearson correlation, $p = 0.02$, permutation test). These results were confirmed with a false-discovery rate correction, and a group-level multilevel model was created, which revealed a significant interaction between the factors congruency and task on gamma-band power ($\chi^2 = 9.2$, $p = 0.002$). In a multilevel model, the authors found an interaction between trial history (incongruent trials preceded by congruent trials vs. incongruent) and gamma power ($\chi^2 = 4.4$, $P = 0.03$), consistent with the Gratton effect, which was stronger in the Stroop task ($p < 0.001$) than the Reading task ($p = 0.72$). In contrast to Sheth *et al.* (2012) (Sheth *et al.*, 2012), Tang and colleagues (2016) did not observe a significant difference in congruent trials preceded by congruent trials vs. incongruent trials. The Gratton effect was present in all studied brain regions with no differences between regions ($F = 0.25$, $p = 0.86$, ANOVA). Regarding the temporal dynamics, the authors showed that latency of modulatory response varied across brain regions ($p = 0.01$, pairwise permutation test with multiple comparison correction) with the ACC activating first, followed in order by the DLPFC, MFC, and OFC (Fig. 4).

The authors also examined other frequency bands in a multilevel model and found that there was an interaction between the congruency and task factors for the theta ($p < 10^{-5}$) and beta (9–30 Hz, $p < 10^{-4}$) bands, which was more prominent in the Stroop than in the Reading task. As opposed to gamma power, theta-band power decreased during incongruent trials and did not correlate with reaction times. They observed significant CFC between theta and gamma oscillations in 50% of electrodes, but unlike Oehr and colleagues (2014) (Oehr *et al.*, 2014), they did not observe a statistically significant difference between congruent and incongruent trials with respect to CFC ($p = 0.52$, sign-rank test). Finally, the authors investigated the neural mechanisms of error processing by looking at the small number of error trials with self-correction (2 patients with greater than 5 error trials each). They found that there was an increase in gamma power after an erroneous response that was not present after the corrected response ($p < 0.05$, sign-rank test), indicating that error detection is signaled through gamma oscillations. Taken together, the results of Tang and colleagues (2016) offered more support for the importance of gamma-band modulation in prefrontal structures in conflict processing and showed that there is some task-specificity in neural signals.

Bartoli *et al.* (2018) investigated gamma- and theta-band oscillations

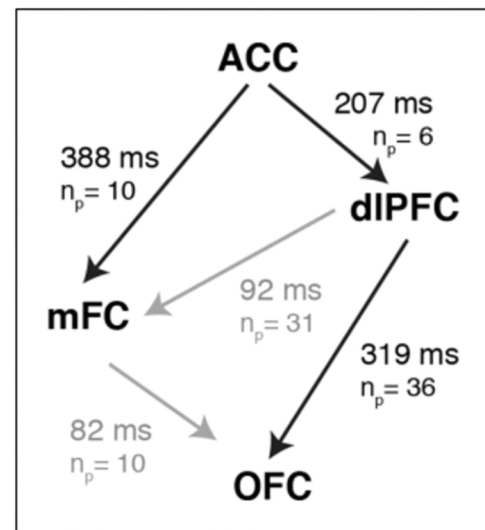


Fig. 4. Latency Comparisons across regions. Latency differences between different regions computed from all pairs of simultaneously recorded electrodes. n_p denotes the number of electrode pairs. Because we only consider simultaneously recorded electrodes here, not all the electrodes modulated by conflict can be paired with any other electrode. Supplementary file 3 shows the number of electrodes modulated by conflict in each area and subject. There was only one electrode pair between ACC and OFC and therefore we do not show the latency difference between these two regions here. Significant latency differences ($P < 0.05$, permutation test, Materials and methods) are shown in black, and non-significant differences in gray. ACC leads both mFC ($P = 0.001$) and DLPFC ($P = 0.02$), with OFC following DLPFC ($P = 0.009$). Reproduced from Tang *et al.* (2016) (Tang *et al.*, 2016).

in the ACC and frontal regions during the color-word Stroop task from SEEG electrodes ($n = 7$ patients) and ECoG electrodes ($n = 6$ patients) in 13 participants undergoing seizure localization (Bartoli *et al.*, 2018). In a stimulus-locked analysis, they found that 70% of contacts ($n = 58$ of 83) had increased gamma-band power ($p < 0.05$, $> 20\%$ increase compared to baseline for 150 ms, permutation test). Results were similar in a response-locked analysis. The bilateral DLPFC and precentral cortex showed strong gamma-conflict modulation (74%, 90%, 50%, 76% of contacts in the middle frontal gyrus, inferior frontal sulcus, superior frontal sulcus, and precentral gyrus, respectively). On an individual level, the proportion of electrodes showing gamma-band modulation included 100% of electrodes in the left superior frontal gyrus, 33% of electrodes in the inferior frontal gyrus, and 67% of electrodes in the cingulate sulcus (all left). They also found that theta power increased in incongruent trials in the cingulate cortex (56%). Temporally, the increase in theta power coincided (725 vs. 739 ms from stimulus onset) with the increase in gamma power in the cingulate cortex but was longer lasting in the gamma band (495 ms vs. 182 ms). At a group level, they also found that gamma power was higher in the incongruent trials than the congruent trials in the DLPFC (34.25% vs. 22.07%, $p < 0.001$, permutation test), and the magnitude of the difference in power changed over time, reaching the highest value at 500–750 ms after stimulus and returning to baseline 1.5–2 s after stimulus. The results were similar when looking at pre-central regions. They also found that the “Gratton Effect” was present in the DLPFC but not premotor regions, as the difference in gamma power between incongruent and congruent trials was greater when the trials were preceded by congruent trials. Furthermore, they demonstrated that beta-band power in the DLPFC decreased following an incongruent trial (DLPFC: 3.01% vs. 8.41%, $p = 0.048$). In a connectivity analysis, they observed that the cingulate cortex had an increase in connectivity to the DLPFC in incongruent vs. congruent trials in the gamma-band ($p < 0.05$, permutation test), and they found an increase in intra-cingulate connectivity in the incongruent condition (p

< 0.05, permutation test). Similar results were obtained when looking at the theta-band. This work further supported the importance of gamma-, theta-, and beta-band activity in prefrontal regions and the ACC in conflict processing.

Smith and colleagues (2019) evaluated conflict processing during an MSIT using LFP and single neuron recordings from the dACC and DLPFC in 6 patients with SEEG for seizure localization, 8 patients undergoing DBS surgery with microelectrode recoding and a subdural strip, and 1 patient undergoing epilepsy monitoring with a prefrontal subdural grid and DLPFC Utah array (Smith et al., 2019). They used a generalized linear model (GLM) with three factors: decision conflict, response identity, and feedback-valence, and they focused on the post-cue period. In the dACC ($n = 136$ neurons), 10.3% of neurons encoded decision conflict, 8.8% encoded response identity, and 8.1% encoded feedback valence, with little overlap between the three groups. 36.8% of dACC neurons had spike-field coherence in the theta-range, and 31.6% had SFC in the beta range ($p < 0.005$). The dACC exhibited phase-coding and experienced neuronal firing before the LFP trough in high conflict trials and after the trough in low conflict trials, and the number of neurons with phase-coding was higher than those with rate coding. Their analysis on spike-triggered LFPs (stLFPs) locally in the dACC revealed that there was a maximal negative deflection in the LFP 100 ms after the spike, and that the stLFP amplitude was greater for rate-encoding than for beta- or theta-coherent neurons. SFC analysis showed temporal coding between dACC beta- and theta-coherent populations and DLPFC LFPs, and this coherence was greater for theta (36%) than for beta (31.6%) frequencies. In addition, the subpopulation of dACC neurons that were coherent with dACC LFPs was distinct from those coherent with DLPFC LFPs. The authors found that dACC stLFPs in the DLPFC was greatest for rate-coding neurons. In the DLPFC ($n = 367$ neurons), the GLM revealed that 4.1% of neurons encoded conflict, 4.8% encoded response identity, and 6.5% encoded feedback valence. The proportion of neurons encoding conflict was smaller than for the dACC and was not significantly greater than expected by chance. The majority of neurons in the DLPFC (64.9%) showed theta-range SFC, which was stronger with higher levels of conflict. This was seen in 52% of individual neurons and in all 9 subjects. This conflict-modulated SFC amplitude scaling was not as prominent in the dACC neurons as in the DLPFC neurons ($p < 10^{-5}$, Fisher's exact test). In contrast, the decision conflict phase-coding observed in the dACC was not as prominent in the DLPFC. There was a precession of SFC phase in all conditions with no difference among conditions. The DLPFC and dACC also showed temporal coding, but the schemes differ between the two structures. Finally, the authors demonstrated that theta SFC in the DLPFC increased in individual trials with higher levels of conflict, and the duration of theta SFC predicted reaction time in each trial (Linear Mixed Model, $t_{886} = 2.9$, $p = 0.002$). All other effects in the model (e.g., current and previous trial conflict) were weaker than the effect of theta-coherence in predicting reaction time. The work by Smith and colleagues (2019) provided important insights into the single neuron activity in the dACC and DLPFC in response to conflict and suggested a mechanism by which different activity in the two structures may relate to their different roles in conflict processing.

In a subsequent paper by the same group, Ebitz and colleagues (2020) sought to model the behavior of single neurons in the dACC and DLPFC in conflict processing during performance of an MSIT task (utilizing the Eriksen "flanker" and Simon tasks) in 16 participants either with SEEG contacts in place for seizure localization or who were undergoing awake DBS placement with microelectrode recording (Ebitz et al., 2020). In the population of dACC neurons examined ($n = 145$, 6 patients), spike activity was higher in the Eriksen task than with no conflict ($p < 0.03$, mean increase = 0.022 z-scored spikes, t-test). 12 out of 145 individual neurons (8.3%) had different activity levels in the Eriksen task than during no conflict, which was greater than chance ($p < 0.003$, binomial test). The Simon conflict had a tendency toward demonstrating increased firing at the population level, but it was not

statistically significant. 15 out of 145 neurons (10.3%) had different firing rates in the Simon conflict, which was greater than chance ($p < 0.003$, binomial test). At the population level, the greatest effect was when both Simon and Eriksen conflict were present, and this fit an additive model better than other models (all BIC weights < 0.02; sig. additive term: $\beta_1 = 0.033$, $p < 0.003$). In the population of DLPFC neurons ($n = 378$ neurons in 9 patients), there was no difference in firing in the Eriksen test, but there was a small but significant effect of firing in the Simon task ($p < 0.005$, mean increase 0.0005 z-scored spikes/s). The number of individual neurons barely exceeded the expected false positive rate in the DLPFC. Fewer neurons in the DLPFC (7.9%) encoded either form of conflict compared to dACC (17.9%) ($p < 0.005$).

The authors next tested if neurons encode abstract or task specific information based on the conflict type (Simon vs. Eriksen). They found that dACC and DLPFC neurons responsive to Eriksen conflict had little overlap with neurons responsive to Simon conflict. They then tested whether the neural activity in the dACC and DLPFC was consistent with the epiphenomenal hypothesis (co-activation of neurons that are tuned for different actions), the explicit hypothesis (dACC neurons signal conflict abstractly to downstream structures e.g. DLPFC), or the amplification hypothesis (conflict amplifies task relevant information at the expense of irrelevant information). They used targeted dimensionality reduction to identify coding dimensions of population activity. Using multiple logistic regression, they projected neural activity associated with specific correct responses into a coding dimension. From this, they were able to predict current correct responses from neural activity. Classification accuracy was higher for trials with Eriksen conflict. They found that the model was most consistent with the amplification hypothesis. By examining the representational geometry of task variable and conflict coding dimension, they sought to answer if there was a task-invariant, abstract conflict coding vector that could decode the presence or absence of conflict, but they did not find this to be the case; rather, there are domain-specific dimensions for different tasks. This work demonstrated the task-specific processing performed by the neurons in the dACC and DLPFC and suggested a possible model as to how this processing occurs.

In another study leveraging single neuron recordings, Fu et al. (2022) recorded LFPs and single unit recordings from the dACC and pre-SMA during the MSIT and the color-word Stroop task (Stroop: $n = 584$ neurons in dACC, $n = 607$ neurons in pre-SMA in 34 participants; MSIT: $n = 326$ in dACC and $n = 412$ in pre-SMA in 12 subjects, Fig. 5A) (Fu et al., 2022b). They used Bayesian inference to model the participants' performance, and the decision process was modeled as a drift-diffusion process. The authors found that conflict was associated with prolonged reaction times, and high estimated conflict probability prolonged reaction time in non-conflict trials while hastening reaction time in conflict trials. They isolated multiple independent populations of neurons encoding: the mean (25% in MSIT, 12% in Stroop) and variance (21% in MSIT, 12% in Stroop) of conflict probability in the ex-post epoch (after action); conflict probability at baseline (21% in MSIT, 17% in Stroop); conflict in the ex-ante epoch (between stimulus and action, 17% in MSIT, 14% in Stroop); conflict probability in the ex-post epoch (21% in MSIT, 20% in Stroop); conflict surprise (16% in MSIT, 12% in Stroop); and errors in the ex-post epoch (20% in MSIT, 30% in Stroop). Neurons signaling conflict in the ex-post epoch was a novel finding in this study. Some neurons "multiplexed" signals (signaled errors, surprise, conflict probability) and increasing proportions of neurons showed multiplexing behavior in later epochs (e.g., ex-post > ex-ante). Neurons encoding conflict probability had greater self-similarity than other neurons with respect to trial-by-trial baseline spike rates, and the extracellular spikes had narrower width compared to other neurons.

The authors then used neuron recordings to train temporal decoder models for error, conflict, conflict probability, and previous trial conflict. Error decoders trained in the early ex-post epoch did not generalize to earlier epochs as well as those in the late ex-post, indicating that some

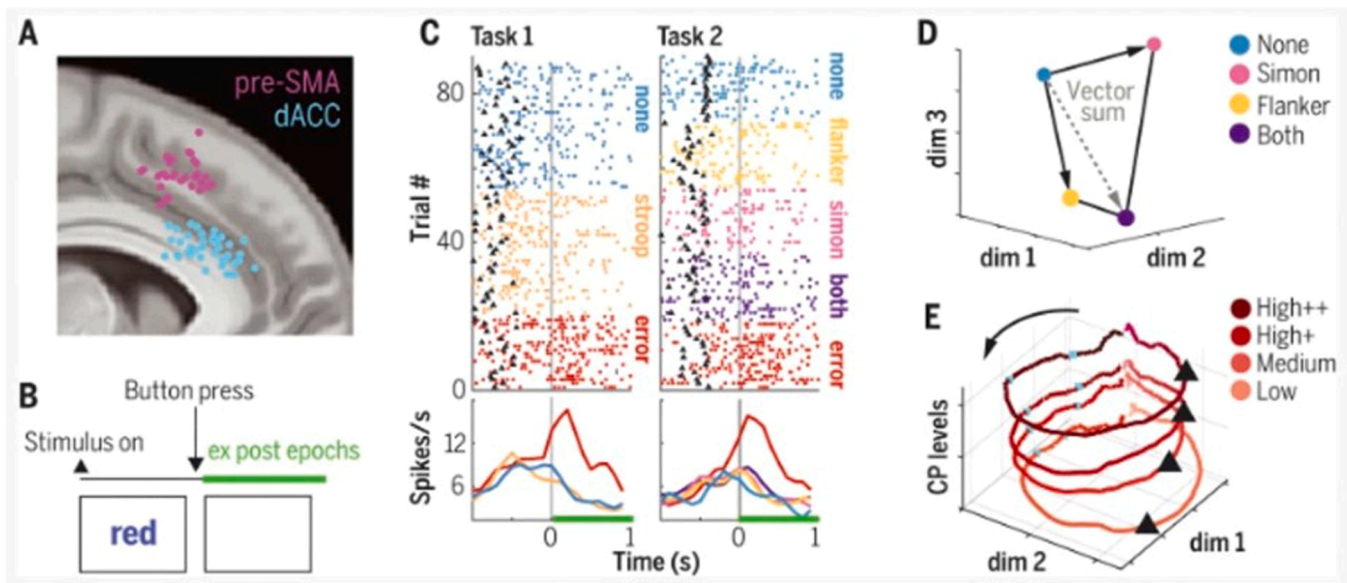


Fig. 5. Representations of evaluative signals in the human frontal cortex are both abstract and task-specific. (A) Recording locations. (B) Analysis epochs. (C) Response of example neuron in both tasks, demonstrating domain generality for errors (red) and no differentiation of different types of correct conflict trials (all other colors). (D) Compositionality of population-level conflict representations. (E) Conflict probability displaces dynamics in neural state space. Reproduced from [Fu et al. \(2022a\)](#) ([Fu et al. 2022a](#)).

neurons have a role in post-error adjustments. Conflict did not generalize well between the ex-ante and ex-post epochs, indicating that conflict processed in these different epochs was done by distinct subpopulations of neurons. Decoders of conflict probability before updating generalized well to those after updating, indicating that conflict probability was stably maintained over time. Models using information from previous trial conflict was weak, especially compared to estimated conflict probability.

They then characterized the representational geometry of conflict in principal component analysis (PCA) space for the MSIT task since it includes four different conflict conditions (Simon, Flanker, Both, and non-conflict), using PCA and decoders trained to differentiate between conditions. From this analysis, they found that they could reliably separate the conditions from each other with high accuracy. The four conditions form a parallelogram in PCA space, and the ‘both’ condition was and should be expected to be located at the linear vector sum of ‘Simon’ and ‘Flanker’ ([Fig. 5D](#)). Using this information, they were able to train decoders that could use one edge of the parallelogram to decode the other edge with high accuracy ($p < 0.001$, permutation test). They also mapped conflict probability (binned into 4 levels) into full neural state space and found that the variability in conflict probability could be depicted in one axis (PC3) that was orthogonal to time-dependent and condition-independent firing rate changes (PC1 and 2, respectively). Conflict probability can be viewed as a state that is present before stimulus onset that returns to baseline after trial completion (depicted by circular motion, [Fig. 5E](#)).

Using demixed PCA (dPCA) at the neuron population level, the authors then sought to demonstrate whether representational geometry could separate between different tasks (Stroop vs. MSIT). From this analysis, they found that there was a task-invariant dimension that was orthogonal to the dimension for task identity when comparing Simon to Stroop and Flanker to Stroop for error, conflict, and conflict probability. The task-invariant decoding dimension could be used to separate different task identities. Finally, they compared the distributions of neuronal selectivity between the dACC and pre-SMA, and they found that the proportions of task-invariant and task-dependent neurons were similar with respect to conflict, conflict probability, and error. There were, however, key differences between the structures with respect to decoding accuracy (higher in the pre-SMA) and timing of task-invariant

error response (earlier in pre-SMA by 0.5 s for Stroop and MSIT). Ex-ante conflict information was observed in the dACC 138 ms before the pre-SMA, and ex-post conflict information was available in the pre-SMA 161 ms before the dACC. In summation, the results from this study clearly defined the roles of populations of single neurons in the dACC and pre-SMA in processing different types of conflict and error and modeled the representational geometry of this processing.

In an interesting study by [Xiao et al. \(2023\)](#), investigators evaluated LFP recordings from SEEG contacts from numerous locations in 16 patients undergoing seizure localizations during the Stroop, Flanker, and MSIT tasks in order to compare results from different types of conflict, focusing their analysis on the theta- and high gamma-bands ([Xiao et al., 2023](#)). They observed that 19% of electrodes showed conflict modulation in at least one task in the high-gamma band, and 16% of electrodes showed modulation in the theta-band. The high-gamma modulation correlated with reactions times in incongruent trials in the three different tasks. In addition, the authors found within-task invariance (i. e., no difference in modulation between different color-word combinations within the Stroop task) for the three tasks. They also used a machine learning decoding approach with a support vector machine (SVM) classifier with a linear kernel. This classifier considered two neural features: the maximum and the mean band power in either the high gamma- or theta-bands. They trained the model with neural responses for several conditions and then tested them on the remaining conditions, and the classifier was able to extrapolate the identity of the trial as congruent vs. incongruent. The majority of individual electrodes that showed modulation in conflict conditions encoded conflict in only one task but not the other two (88% in high gamma and 84% in theta). 12% and 16% of conflict modulating contacts showed modulation in two tasks in the high-gamma and theta bands, respectively. No electrodes showed modulation in all three tasks. They also found that other frequency bands showed conflict modulation in only one task, such as alpha (88%), beta (94%), and low gamma (93%). The authors then trained and tested the linear SVM classifier models using data from the three different populations of electrodes to assess the ability of the models to differentiate between the three different tasks. The Stroop-only population had better accuracy when trained and tested on the Stroop task ($p < 0.001$, permutation test), the Flanker-only population had better accuracy when trained and tested on the Flanker task

($p < 0.001$), and the MSIT-only population had significant classification performance when trained and tested on the MSIT task ($p < 0.001$) as well as on Stroop ($p < 0.01$); however, the Stroop performance was worse than that of the MSIT in this model ($p < 0.001$). This work demonstrated that at the LFP level, populations of neurons in different brain regions encode task-specific information.

Our group [Tang et al. (2021)] investigated the interaction between the OFC and hippocampus by sampling neural recordings from nine patients implanted with SEEG contacts for seizure localization during a modified color-word Stroop task (Tang et al., 2021a). In 6 out of 7 patients analyzed, stimulus-locked theta coherence decreased between the hippocampus and OFC 100–400 ms after cue onset only for successful incongruent trials ($p < 0.05$, cluster permutation test) (Tang et al., 2021a). One patient showed increased theta coherence during the incongruent condition. Group level statistics were computed using a two-way ANOVA to test the effects of conflict, ipsilateral vs. contralateral placement, and the interaction between conflict and sidedness on theta coherence. There was a significant main effect from conflict ($F_{(1599)} = 15.64$, $p < 0.001$). Sidedness and the interaction between conflict and sidedness were not significant. A separate group level analysis looking at the difference in theta coherence values in the incongruent and congruent trials was performed for stimulus- and response-locked segments. The mean z-scores for coherence difference were -1.06 (stimulus-locked) and -0.32 (response-locked) ($t_{(3999)} = -59.81$, $p < 0.001$). Only 11 failure trials were available for analysis. The results in this study are consistent with a study by Young et al. (2011) (Young and Shapiro, 2011b), who observed decreased coherence between the OFC and hippocampus in rats during a novel condition in a maze experiment, while they had increases in theta coherence in remembered situations.

In a subsequent study [Chen et al. (2022)], we recorded LFP activity from SEEG contacts during a modified color-word Stroop task to examine the intra-structural PAC pattern within the OFC in 8 epilepsy patients. We found increased theta-low gamma PAC 200–800 ms after cue presentation in 5 out of 8 patients during successful trials of the incongruent condition ($p < 0.05$, cluster permutation) (Chen et al., 2022) (Fig. 6). At the group level, there was a statistically significant difference between congruent and incongruent theta-low gamma PAC in the OFC during the cue-processing period ($t_{(7999)} = 37.12$, $p < 0.001$,

Cohen's $d = 0.51$, paired t-test). Comparing the response-locked with the stimulus-locked results, there was a statistically significant difference ($t_{(7999)} = 24.80$, $p < 0.001$, Cohen's $d = 0.38$, paired t-test). The theta-low gamma PAC we observed is thought to reflect successful conflict processing.

In a recent study by Herman et al. (2023) (Herman et al., 2023), the authors recorded single unit activity from the dACC ($n = 138$ neurons) and DLPFC ($n = 378$ neurons) during an MSIT task in 10 patients undergoing SEEG for seizure localization and 8 patients undergoing DBS (7 for Parkinson's Disease, 1 for epilepsy) (Herman et al., 2023). They sought to determine if pre-trial firing rates of neurons could predict reaction time in the subsequent trial. Using a generalized linear mixed effects modeling (GLME) approach, they found that a conflict-type general model fit dACC activity ($p < 10^{-43}$, Benjamini-Hochberg corrected) while DLPFC neurons fit a conflict-type general model ($p < 0.003$, Benjamini-Hochberg corrected). Firing rate in the dACC was associated with response time in the presence of any conflict ($F = 6.6$, $p = 0.036$, Bonferroni corrected permutation test). In contrast, the DLPFC showed a conflict-type specific association between firing and response time ($F = 13.5$, $p < 0.001$). To evaluate the temporal relationship between dACC and DLPFC activity and reaction time, the authors performed a sliding window GLME. From this analysis, they found that the peak time for the association between dACC conflict-type general coding and response time occurred 1500 ms before stimulus onset ($F = 10.97$, $p < 0.001$, permutation test), whereas the peak association between DLPFC conflict-type specific coding and response time occurred 750 ms before the cue ($F = 12.2$, $p = 0.0016$, permutation test). They then calculated GLM vector regression weights to Simon and Ericksen conflict conditions for the DLPFC and found that coding for Simon and Ericksen conflict were orthogonal to each other in the period preceding cue presentation ($\theta = 98.4^\circ$) as well as in the response phase ($\theta = 92.5^\circ$). In contrast to the DLPFC, the coding for Simon and Ericksen conflict were partially colinear for the dACC in the preparatory ($\theta = 63.2^\circ$) and the response periods ($\theta = 75.5^\circ$). Finally, they demonstrated that the coding in the preparatory state is orthogonal to the coding of the response state for the Simon ($\theta = 87.2^\circ$) and Ericksen ($\theta = 88.1^\circ$) conditions in the DLPFC, while it is partially co-linear for the dACC for both types of conflict ($\theta = 50.6^\circ$ for Simon vs. $\theta = 66.9^\circ$ for Ericksen), indicating that the dACC has complementary mechanisms for processing conflict.

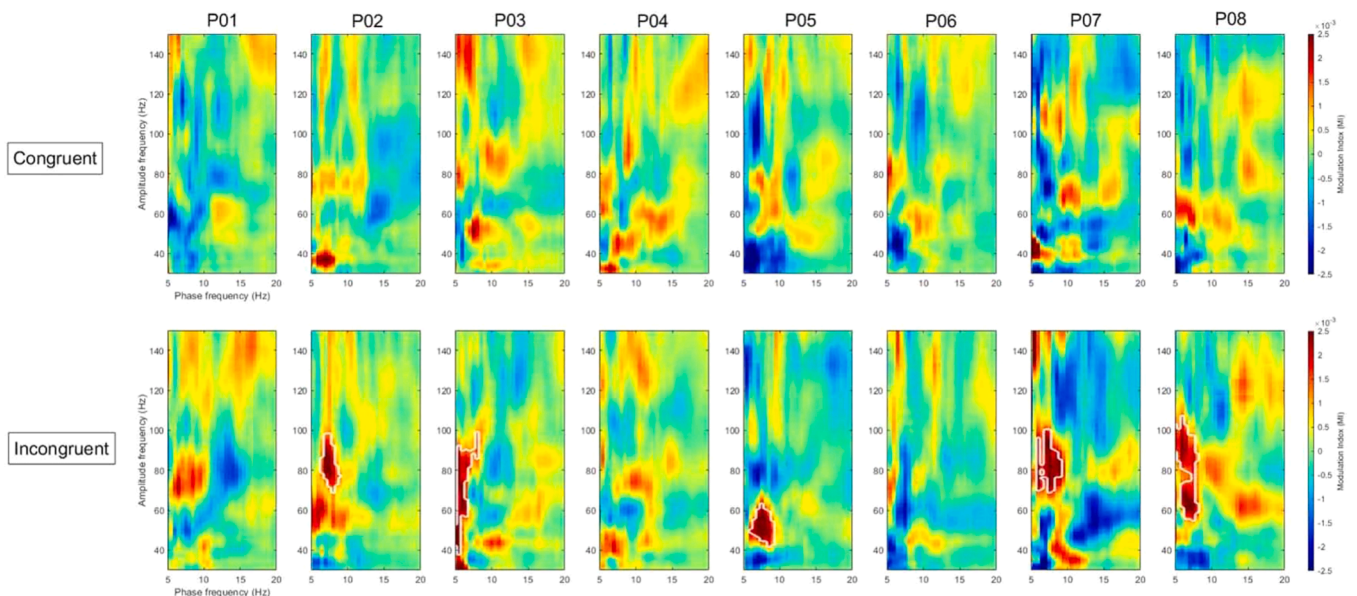


Fig. 6. Increased theta-LG PAC values in the OFC during successful trials in the incongruent condition of our modified Stroop Task (Chen et al. 2022). This figure displays trial-averaged comodulograms for the difference between the cue-processing period and baseline with each column representing a patient. The upper row shows the plots for congruent conditions, while the bottom row shows for incongruent conditions. The overlaid white polygons Kullback–Leibler (KL)-based modulation index points of significant PAC changes compared to baseline. This figure was adopted from Chen et al. (2022). (Chen et al. 2022).

3. Hippocampus

The hippocampus is most classically known for its roles in memory and learning (Opitz, 2014; Lazarov and Hollands, 2016). More recently, a rapidly increasing number of human physiology studies have also begun implicating the hippocampus in conflict processing (Bach et al., 2014; O’Neil et al., 2015; Çavdaroglu et al., 2021). Initial studies that combined various forms of the Stroop paradigm with a broad use of fMRI delivered the first glimpse into hippocampal involvement during this specific type of cognitive processing (Ramm et al., 2020a; Ramm et al., 2020b; Ramm et al., 2021; Spaniol et al., 2009). Ramm et al. (2020) (Ramm et al., 2020a) demonstrated that patients with hippocampal sclerosis less accurately performed in a Stroop task compared to healthy controls, indicating a likely functional involvement of the hippocampus in conflict processing. However, one study investigating Stroop performance before and after unilateral temporal lobectomy did not show a difference when comparing pre- and post-operative performance (Güvenç et al., 2018). Overall, these studies paint a broad picture of hippocampal activation in the processing and resolution of conflict.

SEEG is commonly used to evaluate hippocampal involvement in patients with medically refractory epilepsy and therefore represents a valuable opportunity to study neural oscillatory changes during conflict resolution. Oehrns et al. (2015) recorded data from fMRI and SEEG (termed “intracranial electroencephalography” or “iEEG” by the authors) in 10 patients with pharmacoresistant epilepsy to investigate hippocampal activation using an auditory version of the Stroop task, wherein the pitch and the meaning of sounds “high” and “low” are used instead of color-word combinations (Oehrns et al., 2015). When analyzing and processing stimulus-locked hippocampal theta-band power of incongruent trials relative to congruent trials, they observed decreases in theta power approximately 2 seconds after cue presentation in 9 patients (4–7 Hz, $p = 0.02$, cluster permutation test). (Fig. 7). Furthermore, response-locked oscillations showed an increase in theta power ($p = 0.046$, cluster permutation test) for incongruent trials relative to congruent trials that occurred 900–400 ms before the verbal response and then a subsequent decrease ($p < 0.01$, cluster permutation test) 520–1500 ms after the verbal response. The observed changes in theta power were specific to the hippocampus and not seen in the rhinal cortex or temporobasal cortex. In an analysis of correct vs. incorrect incongruent trials, they found increased pre-response theta power (3–7 Hz, –866 to –105 ms, $p < 0.01$, cluster permutation test), increased pre-response gamma power (76–152 Hz, –287 to –248 ms, $p = 0.03$, cluster permutation test), and an increase in post response

broadband gamma power (45–181 Hz, 361–431 ms, $p < 0.001$, cluster permutation test) in correct trials. The post-stimulus and post-response theta power decreases are thought to mirror the same change as reaction times on average for the tasks in this study, which were around 1000 ms. Interestingly, pre-response increases in low-gamma (32–45 Hz) power were also found during response-locked analysis (–151 to –106 ms, $p = 0.049$, cluster permutation test), suggesting that pre-response hippocampal theta and low gamma increases could potentially be related to preparation of the verbal response. Consistent with the intracranial recordings, there was increased BOLD signal on fMRI in the incongruent condition compared to the congruent condition (Z-core 4.26, $p < 0.008$, ANOVA). Interestingly, they observed increased BOLD signal in the inferior frontal gyrus during the incongruent condition as well (Z-score 3.24, $p < 0.033$, ANOVA).

While Oehrns et al. (2015) characterized hippocampal theta and gamma changes during conflict processing, our group [Chen et al. (2020)] sought to examine the beta frequency band in six patients with SEEG implanted for seizure localization (Chen et al., 2020). We used a classic color-word Stroop paradigm to evaluate changes in beta band power during incongruent trials of the Stroop task (Fig. 8). Although some patients in our study had bilateral lead implants with both right and left hemisphere contacts, only right hemisphere contacts were appreciated to be active during the Stroop task. We found statistically significant increases in beta-band power in the incongruent condition that occurred 500–1000 ms after cue presentation (stimulus-locked) in five out of six patients ($p < 0.05$, cluster permutation test) (Fig. 9). Similar findings were reported in response-locked analysis (Fig. 10), though specific timestamps were more variable between 1000 ms prior to the verbal response up to the verbal response itself. Beta power increases for the incongruent condition were more prominent in the response-locked analysis compared to the stimulus-locked one. No beta-band changes were noted during congruent trials. In a group level analysis (paired sample permutation t-test), there was a statistically significant difference (mean z-score 2.64 for response locked and 1.19 for stimulus locked, $t(3999) = -68.21$, $p < 0.001$).

4. Amygdala

The amygdala is a deep brain structure that is part of the limbic system. It has well-documented activity in processing emotional conflict during fearful learning tasks (Weiskrantz, 1956; LeDoux et al., 1990; Blanchard and Blanchard, 1972; Anderson and Phelps, 2001); generating excitatory responses to previously conditioned olfactory, auditory,

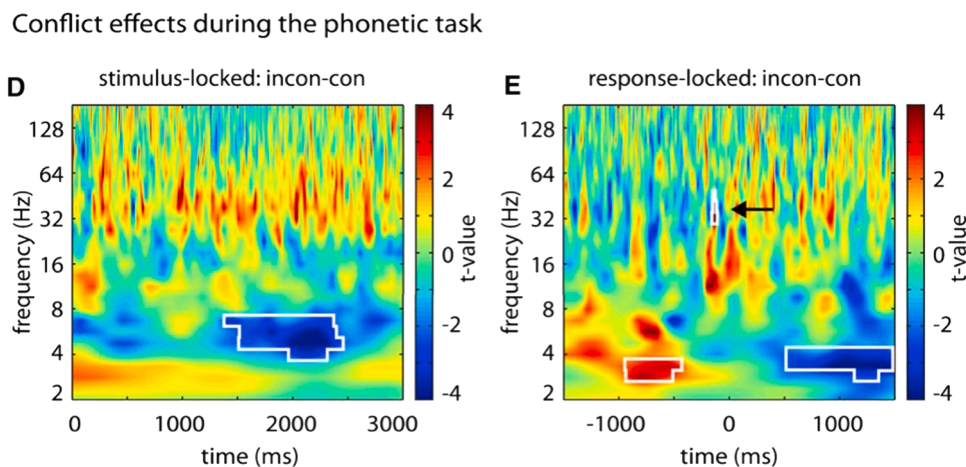


Fig. 7. Analysis of hippocampal power data during a phonetic Stroop task designed by Oehrns et al. (2015). These graphs chart the color-coded time-frequency resolved test statistics comparing power values during both correct-inconsistent and consistent stimulus processing. (D) represents the stimulus-locked analysis where zero marks stimulus onset. (E) represents the response-locked analysis where zero marks the response onset. This figure was adopted from Oehrns et al. (2015). (Oehrns et al. 2015).

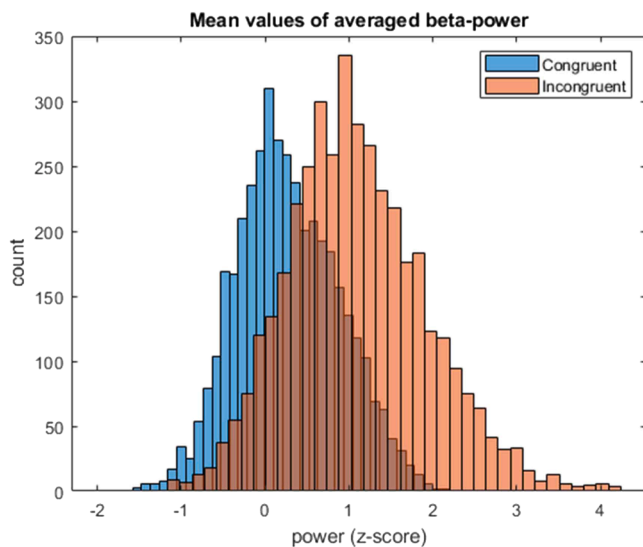


Fig. 8. Hippocampal mean beta power increases during our modified Stroop Task in 6 patients with medically refractive epilepsy. This figure displays the overlaid peak beta-power range during both congruent and incongruent conditions. This figure was adopted from [Chen et al. \(2020\)](#). (Chen et al. 2020).

and visual stimuli (Sanghera, Rolls and Roper-Hall, 1979; Chiba, Schoenbaum and Gallagher, 1998; Uwano et al., 1995); suppressing feeding behavior (Cai et al., 2014); and resolving emotional incongruency using face-emotion Stroop testing (Etkin et al., 2006; Kim et al., 2004). In 2005, Hsu et al. suggested that the amygdala plays a role in evaluating general uncertainty after observing a positive correlation between the level of ambiguity in decision making and neuronal activation in the amygdala and orbitofrontal cortex (Hsu et al., 2005).

Based on this study, our group sought to determine the amygdala's role in non-emotional conflict processing (Tang et al., 2021b). Nine patients with epilepsy who were implanted with intracranial depth electrodes participated in a modified color-word Stroop task that included four task conditions. After performing this task, we conducted time frequency analysis of LFP data from macro-electrode contacts located in amygdaloid gray matter. Of the 7 patients included in the

analysis, 5 showed significant increases in theta band (4–7 Hz) activity during the cue-processing phase (i.e., post stimulus, pre-response) of the incongruent condition ($p < 0.05$, cluster permutation test) (Fig. 11). During stimulus-locked analysis, increases in theta power were generally seen 500–1000 ms after cue presentation (Fig. 12). Response-locked analysis revealed theta increases beginning 500 ms before the response and remaining until the verbal response was delivered (Fig. 13). One patient showed theta-band increases during both the congruent and incongruent trials. Overall, theta power increases in the incongruent condition were more prominent in response-locked analysis than stimulus-locked [mean Z-scores for stimulus-locked = 1.62 and for response-locked = 1.16, respectively, $t_{(6999)} = -7.98$, $p < 0.001$, Cohen's $d = 0.17$ (small effect size)]. Overall, these findings are consistent with those of [Oehrn et al. \(2014\)](#) in the neighboring hippocampus, indicating that these two functionally and structurally related regions may also share functionality in conflict processing.

5. Subthalamic nucleus

The subthalamic nucleus (STN) is a key node in the response inhibition network. Research utilizing DBS stimulation and fMRI has suggested that the STN rapidly inhibits basal ganglia activity to pause motor output during conflict until an appropriate motor plan is ready ([Filali et al., 2004](#); [Aron and Poldrack, 2006](#)). The Stroop color-word interference task is a well-established part of preoperative DBS neuropsychological assessment ([Strouwen et al., 2016](#)) and the evaluation of motor-related performance after DBS treatment ([Schroeder et al., 2002](#); [Irmén et al., 2017](#); [Ghahremani et al., 2018](#)). The proposed role of the STN in motor inhibition during conflict is reflected in Parkinson's disease (PD) patients who typically perform poorly on the Stroop task ([Bentivoglio et al., 2013](#); [Ruitenberg et al., 2019](#); [Yoon et al., 2014](#)).

[Schroeder et al. \(2002\)](#) evaluated changes in PD patients' regional cerebral blood flow during a Stroop color-word interference task during the on and off states of bilateral STN stimulation ([Schroeder et al., 2002](#)). Patients undergoing STN stimulation had a pronounced Stroop effect (longer response time) that was associated with reduced activation in the right ACC and right ventral striatum. Thus, the performance of the Stroop task provided a metric to evaluate inhibition performance under the ON/OFF switch of STN stimulation, and it provided direct evidence that the STN plays a key role in modulating basal

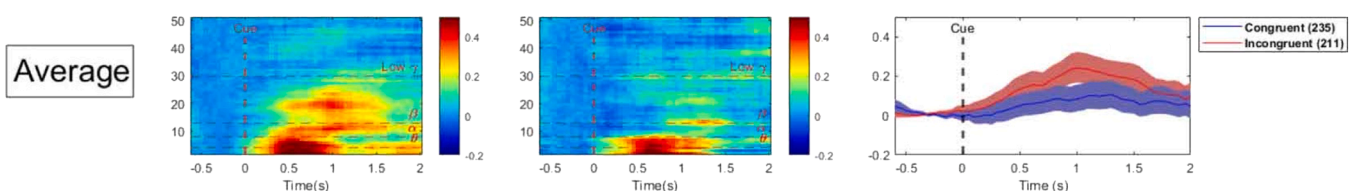


Fig. 9. Stimulus-locked trial-averaged beta-power changes in the Hippocampus. The first column demonstrates the average power spectrogram during the cue-processing period for incongruent trial conditions. The second column demonstrates the average power spectrogram for congruent trial conditions. The third column isolates the beta-band power to show increases that occur during the cue-processing period. The spectrograms in this row represent the averages across all 6 patients included in our study. This figure was adopted from [Chen et al. \(2020\)](#). (Chen et al. 2020).

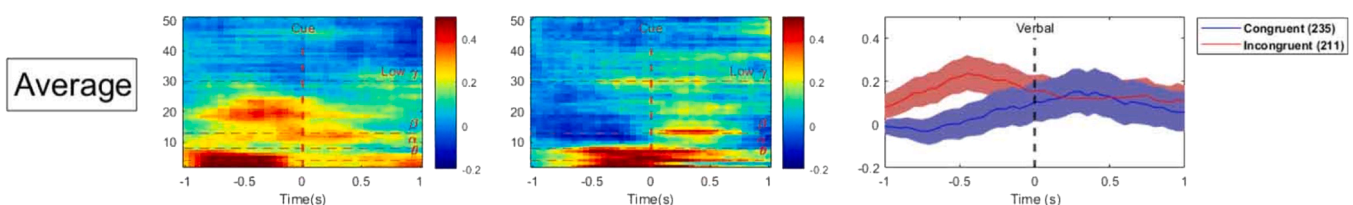


Fig. 10. Response-locked trial-averaged beta-power changes in the Hippocampus. The first column demonstrates the average power spectrogram during the cue-processing period for incongruent trial conditions. The second column demonstrates the average power spectrogram for congruent trial conditions. The third column isolates the beta-band power to show increases that occur during the cue-processing period. The spectrograms in this row represent the averages across all 6 patients included in our study. This figure was adopted from [Chen et al. \(2020\)](#). (Chen et al. 2020).

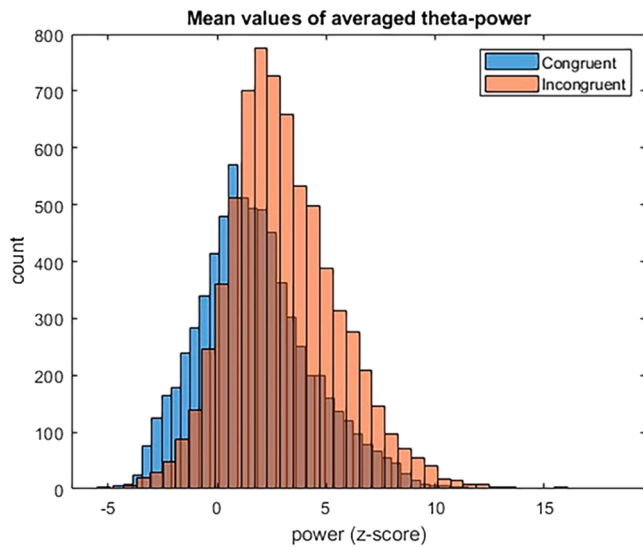


Fig. 11. The mean values of averaged amygdaloid theta power increases during our modified Stroop Task in 7 patients with medically refractory epilepsy. This figure displays the overlaid peak theta-power range during both congruent and incongruent conditions. This figure was adopted from Tang et al. (2021). (Tang et al. 2021).

ganglia-thalamocortical circuitry. Ghahremani et al. (2018) evaluated the Stroop effect of different timed deliveries of event related STN DBS (Ghahremani et al., 2018). The results demonstrated that event-related DBS stimulation of the STN had specific effects on conflict compared to non-conflict trials. Specifically, STN stimulation during the ready period (before the imperative cue) of conflict trials increased the speed of responses. Additionally, when it was delivered early in the response phase of conflict trials, patients had an associated increase in errors.

Ghahremani et al. (2018) further found that low frequency oscillations (2–8 Hz) occurred in the STN during conflict and were observed in

the absence of stimulation during the period after the cue (Ghahremani et al., 2018) (Fig. 14), suggesting that oscillations play a role in conflict processing in the STN. The study utilized the Stroop task to clarify the timing of conflict processing after cue onset by delivering DBS stimulation in an event-related manner. Brittain et al. (2012) similarly used DBS electrodes implanted in the human STN to record neural activity during a Stroop task (Brittain et al., 2012). The authors found increased theta-band (5–10 Hz) activity during the color-word conflict scenario (Fig. 15). In addition, the resynchronization (decreased and then increased) of beta-band (15–35 Hz) activity was found before the verbal response and was therefore recognized as a signal that pauses the motor system until reaching conflict resolution. The timing of this rebound beta activity was shown later in error trials compared to successful trials of the incongruent condition after the verbal response had been made.

6. Discussion

Studies of human conflict processing, particularly those involving the Stroop task, have most extensively targeted frontal cortical regions such as the ACC and PFC using EEG and fMRI. As mentioned previously, these studies formed the basis for the conflict monitoring theory (Botvinick et al., 2001) and the conflict-control loop theory (Carter and van Veen, 2007), which postulates that the dACC acts to detect conflict and then recruits the DLPFC to resolve this conflict. More recently, direct intracranial recordings from humans have added more depth to these analyses and have offered insights into the mechanisms of conflict processing from the level of populations of neurons down to the level of single neurons.

6.1. Cortical Regions

As previously discussed at length, most of the studies involving direct intracranial recordings focused on recording signals from the dACC and other frontal cortical structures. The work by Sheth et al. (2012) (Sheth et al., 2012) was consistent with the conflict-monitoring theory as it demonstrated increased firing of dACC neurons after cue-presentation in

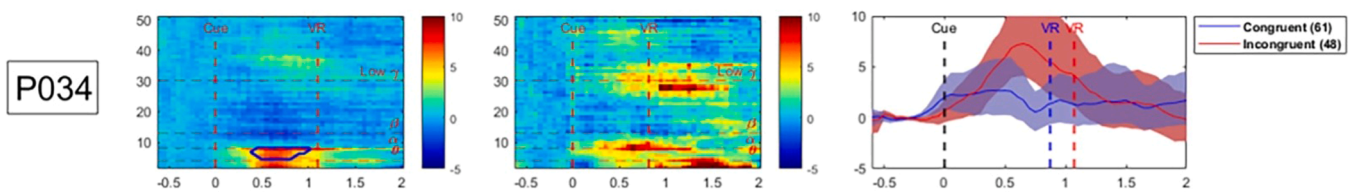


Fig. 12. Stimulus-locked trial-averaged theta-power changes in the Amygdala. This figure highlights a representative patient's (P034) trial-averaged stimulus-locked spectrograms in our study. The first column demonstrates the average power spectrogram for the incongruent trial condition. The periods of significant theta-band modulation are encapsulated within the blue polygon. Meanwhile, the second column shows the average power spectrogram for the congruent trial condition. To better visualize the theta-band power changes for both conditions, the final column isolates mean theta-band power with 95% confidence intervals. The red line marks the trial-averaged mean power change in the incongruent condition. On the other hand, the blue line corresponds to the congruent condition. This figure was adopted from Tang et al. (2021a). (Tang et al. 2021a).

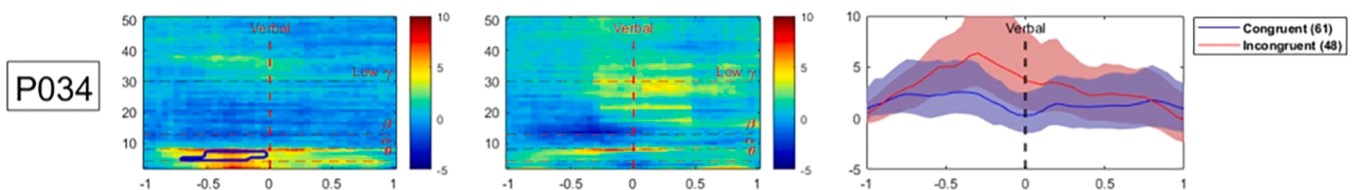


Fig. 13. Response-locked trial-averaged theta-power changes in the Amygdala. This figure highlights a representative patient's (P034) trial-averaged response-locked spectrograms in our study. The first column demonstrates the average power spectrogram for the incongruent trial condition. The periods of significant theta-band modulation are encapsulated within the blue polygon. Meanwhile, the second column shows the average power spectrogram for the congruent trial condition. To better visualize the theta-band power changes for both conditions, the final column isolates mean theta-band power with 95% confidence intervals. The red line marks the trial-averaged mean power change in the incongruent condition. On the other hand, the blue line corresponds to the congruent condition. This figure was adopted from Tang et al. (2021a). (Tang et al. 2021).

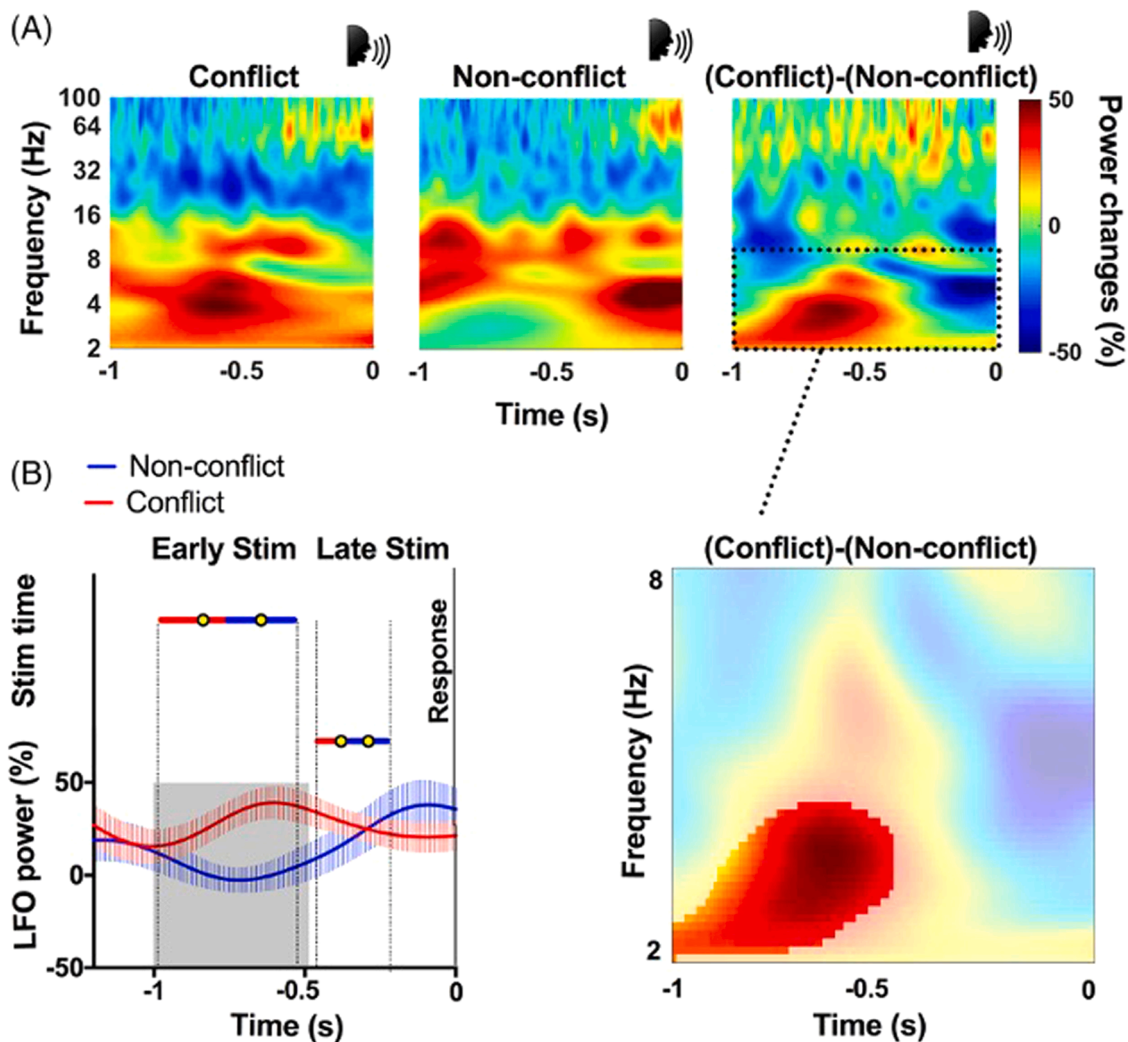


Fig. 14. STN activities during conflict within a Stroop task (Ghahremani et al. 2018). (A) These plots show response-locked power changes across frequencies during both conflict and non-conflict trials. The dotted-box inset highlights a band of interest to examine the contrast of low-frequency oscillations (LFO) between conflict and non-conflict conditions. (B) The timing of stimulation (mean $\pm$ SD) in the pre-response period relative to significant LFO power changes. The vertical lines represent the range of stimulation timing. The grey shaded area represents the significant difference timing between conflict and non-conflict trials extracted from the panel in (A). This figure was adopted from Ghahremani et al. (2018). (Ghahremani et al. 2018).

an MSIT trial, indicating that the dACC plays a role in detecting conflict. Furthermore, the authors showed that conflict-signaling in the dACC is necessary for maintaining a history-dependent modulation of behavioral response (the “Gratton Effect”) that considered interference in the previous trial; moreover, ablation of this structure led to loss of this behavioral modification. The work by Tang et al. (2016) (Tang et al., 2016) supported the conflict-control loop theory, as the investigators found increased gamma power in incongruent Stroop trials in the dACC, DLPFC, OFC, and MFC. The temporal dynamics of response was also consistent with the conflict-control loop theory, in that the dACC was activated first, followed in turn by the DLPFC, MFC, and OFC. They also observed gamma-band power modulation consistent with the Gratton Effect. In keeping with the conflict-control loop theory, Bartoli and colleagues (2018) (Bartoli et al., 2018) demonstrated that gamma and theta power increased in the incongruent condition of a Stroop task in the DLPFC and dACC, and that there was increased gamma connectivity between the DLPFC and dACC in incongruent conditions. Single-unit recordings from Smith et al. (2019) (Smith et al., 2019) and Ebitz et al. (2020) (Ebitz et al., 2020) were also consistent with the conflict-control loop theory, as they both demonstrated an interplay between the dACC and DLPFC. Additionally, single neuron recordings from Herman et al. (2023) (Herman et al., 2023) also lend support to the

conflict-control loop theory, as this study showed that activity in the dACC preceded activity in the DLPFC with respect to firing before the cue presentation.

With respect to LFP analyses, there was significant overlap across studies, most of which investigated the roles of the gamma- and theta-bands. The majority of these LFP studies demonstrated conflict-associated increases in gamma-band power in the DLPFC (Koga et al., 2011; Caruana et al., 2014; Oehrn et al., 2014; Tang et al., 2016; Bartoli et al., 2018), DMPFC (Oehrn et al., 2014), PMC (Koga et al., 2011; Caruana et al., 2014; Bartoli et al., 2018; Xiao et al., 2023), SMA (Koga et al., 2011; Caruana et al., 2014), OFC (Tang et al., 2016), and dACC (Tang et al., 2016; Bartoli et al., 2018). Gamma oscillations convey local information processing and have also been shown to induce adjustments in cognitive control in the dACC and DLPFC (Quilodran, Rothé and Procyk, 2008; Rothé et al., 2011). Several studies also showed a role for theta-band power in conflict-associated modulation. Frontal theta oscillations have been implicated in memory processing, novelty detection, and likely indicate how the need for cognitive control is realized and executed (Cavanagh and Frank, 2014). Bartoli et al. (2018) (Bartoli et al., 2018) showed increased theta-band power in the dACC in the incongruent condition during a conflict task. In contrast, Tang et al. (2016) (Tang et al., 2016) demonstrated a decrease in theta-band power

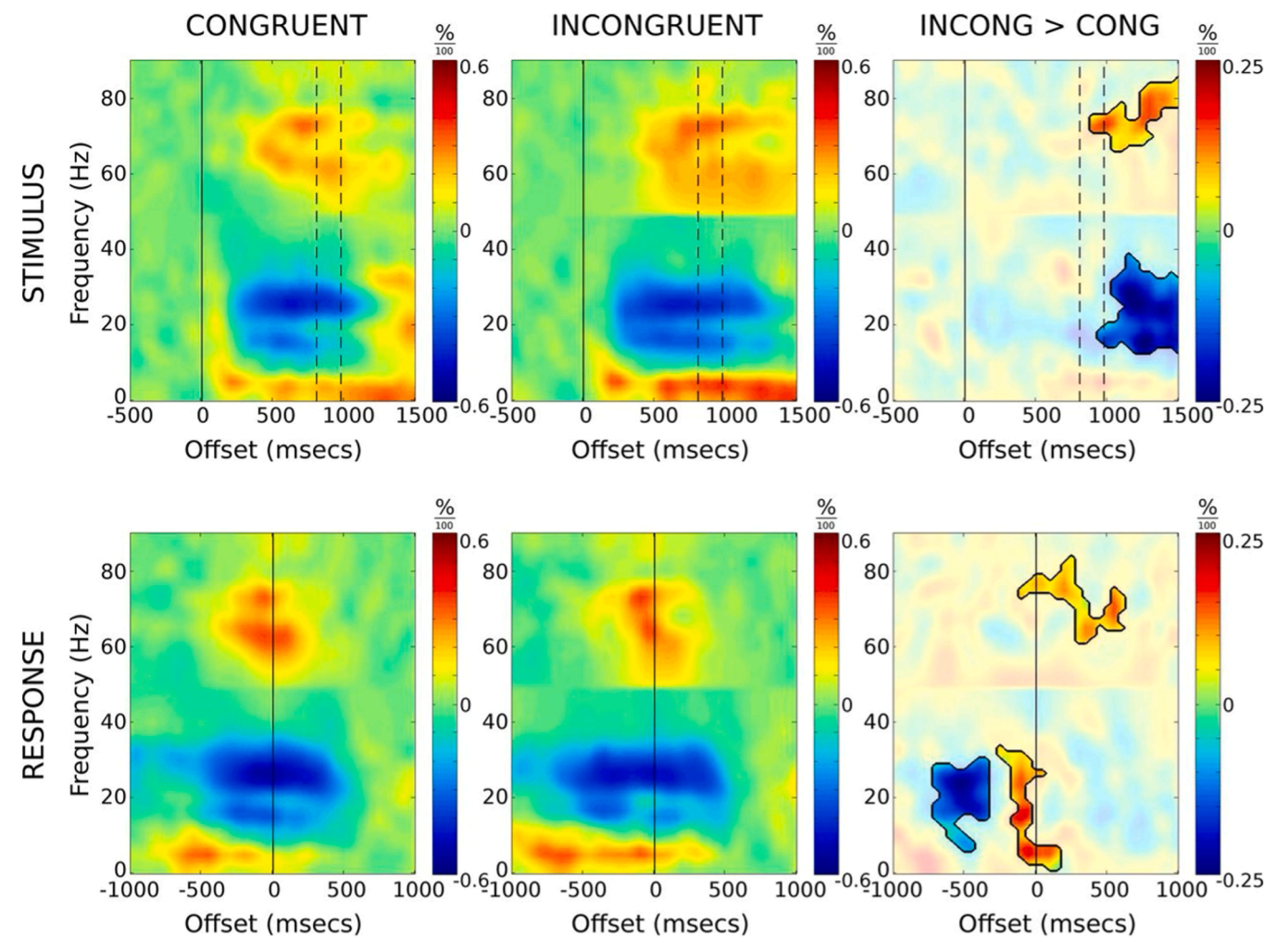


Fig. 15. Stimulus-locked (top row) and response-locked (bottom row) average time-frequency spectrograms for congruent and incongruent trials during a Stroop Task. The third column represents difference spectrograms with incongruent > congruent. The dashed vertical lines represent the mean reaction times for congruent and incongruent trials. This figure was adopted from [Brittain et al. \(2012\)](#). ([Brittain et al. 2012](#)).

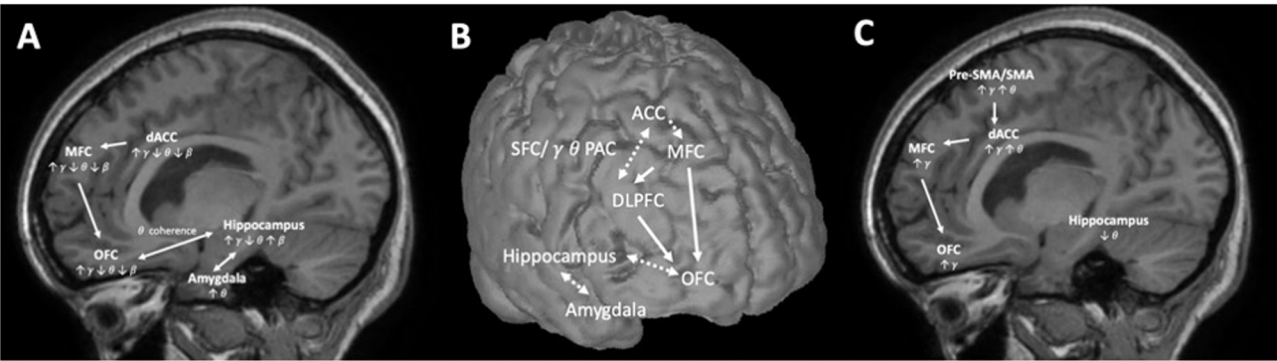


Fig. 16. Network model of human conflict processing. A) model of conflict processing in mesial structures, B) model of conflict processing as seen from the brain surface C) model of error processing in the mesial brain structures. For all panels, solid arrows indicate temporal relationships, relative changes in gamma-, theta-, and beta- band power in the incongruent vs. congruent condition of conflict tasks shown under label for brain region; dotted arrows indicate connections to deep structures not visible at the brain surface. All images derived from a non-contrasted T1 MRI from one of the participants in [Chen et al. \(2022\)](#). ([Chen et al. 2022](#)).

in incongruent conditions. Using LFP data, several studies also evaluated the role of cross frequency coupling in conflict processing. Gamma-theta PAC has been shown to be important in communication across and within brain regions during cognitive processing ([Canolty et al., 2006](#)). [Oehr et al. \(2014\)](#) ([Oehr et al., 2014](#)) showed theta-gamma PAC

between the DLPFC and DMPFC during successful conflict processing in the auditory Stroop task. Similarly, our group demonstrated increased theta-gamma PAC within the OFC in incongruent vs. congruent trials of a Stroop task.

In addition to multi-unit recordings, numerous studies provided a

mechanistic understanding of conflict processing on a single neuron level. For example, Sheth et al. (2012) (Sheth et al., 2012) demonstrated that the cue-responsive neurons in the dACC had increased firing rates with increasing levels of interference. Smith et al. (2019) (Smith et al., 2019) also found that some dACC neurons encoded task (MSIT) conflict by increasing firing rate, but that this effect was absent in the DLPFC. However, the DLPFC did show spike-field coherence, indicating that the dACC detects conflict and recruits the downstream DLPFC for resolution. In recording single-neuron activity in the dACC and DLPFC in an MSIT task, Ebitz et al. (2020) (Ebitz et al., 2020) found that the neuronal populations in the dACC had greater modulation in firing than DLPFC neurons and that conflict amplifies the task-relevant information to enhance allocation of neural resources to the correct response. Using single-neuron recordings, Fu et al. (2022b) (Fu et al., 2022b) modeled the representational geometry of conflict processing in the dACC and pre-SMA and found that single-unit recordings from these structures conveyed both task-specific and domain general information. Herman et al. (2023) (Herman et al., 2023) also showed that increased single-unit firing in the dACC and DLPFC is associated with decreased reaction time.

Error monitoring is another important facet in conflict processing that was studied in frontal cortical regions. Using LFP recordings from the SMA and mesial frontal cortex, Bonini et al. (2014) (Bonini et al., 2014) identified a caudo-rostral axis of error processing in which the SMA was recruited first and most consistently in error monitoring. Similarly, using single-unit recordings, Fu et al. (2019) (Fu et al., 2019) showed that there was a population of error-responsive neurons in the dACC and pre-SMA that signaled within-trial error and error-integrating neurons, which signaled error in the previous trial. Consistent with the findings of Bonini et al. (2014), they found that error signals in the pre-SMA temporally preceded error signals in the dACC. Building off of this work, Fu et al. (2022b) (Fu et al., 2022b) showed that single neurons in the pre-SMA and dACC encode domain-general error signal.

One of the major questions in the study of conflict processing is as follows: is conflict information abstract or task specific? As mentioned earlier in the manuscript, noninvasive recordings have shown differential activation in different types of tasks (Scerrati et al., 2017). Several studies have attempted to address this question using direct neural recordings. Tang and colleagues (2016) (Tang et al., 2016) compared LFP results from a standard color-word Stroop task and a reading task, and they found that gamma power was higher, theta power was lower, and the Gratton effect was more prominent in the frontal structures during incongruent trials in the Stroop as compared to the reading task, perhaps indicating some task specificity. Using single neuron recordings from the dACC and DLPFC during the MSIT, which involves Simon conflict, Ericksen flanker conflict, and the combination of the two, Ebitz et al. (2020) (Ebitz et al., 2020) offered more information on task specificity. By applying a GLM analysis, they found that Simon and flanker conflicts were additive and that a greater proportion of neurons in the dACC was responsive to flanker conflict than Simon conflict. These two pools of neurons also had little overlap, indicating that neurons in the dACC encode task-specific information at the single neuron level. At the multi-unit level, Xiao et al. (2023) (Xiao et al., 2023) found that many different brain regions encode task-specific information (gamma- and theta-band modulation) in ways that differ between the Stroop, Flanker, and MSIT tasks. Although they had relatively fewer frontal contacts, their observations did reveal task specificity in the cingulate, lateral orbitofrontal, medial orbitofrontal, and rostral middle frontal cortices. Our own studies only examined the color-word Stroop task, so we cannot comment on the task-specificity vs. generality of conflict processing in the hippocampus, amygdala, or OFC. However, Xiao et al. (2023) did include some analyses of these structures, and they found that there was task specificity in the amygdala, hippocampus, and OFC in both gamma- and theta-band modulation. Using single unit recordings from the dACC and pre-SMA, Fu et al. (2022b) (Fu et al., 2022b) demonstrated that some neurons encoded task-invariant information while other neurons

encoded task-specific information, and population activity from these neurons could be used to separate different conditions within the MSIT task and different tasks. Herman et al. (2023) (Herman et al., 2023) specifically showed that single-neuron activity in the dACC encodes task-invariant information while activity in the DLPFC encodes task-specific information when looking at the preparatory period before cue presentation. Taken together, the body of evidence from intracranial recordings supports a model in which some neurons code abstract, task-invariant information while other neurons encode task-specific information, but the population level activity of these different neurons (as interpreted by LFPs) conveys task-specific information.

6.2. Other regions

Our group utilized SEEG to find novel neural oscillatory activity in the OFC, amygdala, and hippocampus during the Stroop Task. Strong anatomical connections between these cortical and subcortical structures align with the notion that they cooperate to process cognitive functions, including conflict detection, resolution, and adaptation (Aron et al., 2016; Ray and Zald, 2012; Groenewegen, Wright and Uylings, 1997; Tanaka, Buckley and Mansouri, 2009). In the context of conflict processing, cortical inter-regional interactions have been heavily investigated (Popov et al., 2018; Oehrn et al., 2014; Carter et al., 1998; Botvinick, Cohen and Carter, 2004). However, the role of cortico-limbic communication in these cognitive processes is still not well understood.

6.2.1. Hippocampus and Amygdala

Our hippocampal study showed an incongruent beta power increase that was strongest when locked to the response rather than the stimulus (Chen et al., 2020). In 2012, Brittain et al. observed a similar relationship when analyzing mechanisms of conflict processing in the STN through the utilization of the Stroop task and direct intracranial recordings. They hypothesized that this response-locked beta-band rebound, exclusive to the incongruent condition, might be related to pausing the motor system to resolve conflict (Brittain et al., 2012). Because the hippocampus plays a role in suppressing motor responses to stimuli (Taylor et al., 2014; Kimble and Kimble, 1965), and increases in hippocampal beta power are associated with motor inhibition (Del Campo-Vera et al., 2021), the increase in hippocampal beta-band power observed in our study may be related to motor suppression in the presence of conflict. This proposal is further supported by the fact that beta responses were strongly response-locked, indicating their association with the motor response.

In 2015, Oehrn et al. used SEEG and the Stroop task to determine the role of the hippocampus in processing non-emotional conflict (Oehrn et al., 2015). Specifically, they observed an increase in theta-band power in the hippocampus during the post-stimulus, pre-response period for incongruent trials (i.e., 900–400 ms before the verbal response). Interestingly, our group observed theta power increases in the amygdala during incongruent trials of the Stroop task that begin approximately 1000–500 ms before the verbal response (Tang et al., 2021b). Thus, this increase in amygdaloid theta power appears to occur sequentially to hippocampal theta activity with 100 ms of overlap (i.e., during the 500–400 ms pre-response period) (Oehrn et al., 2015). Previous research has provided behavioral, electrophysiological, and biochemical evidence of cooperative interaction between the hippocampus and amygdala in other functions including the formation of long-term memory (Yang and Wang, 2017). Moreover, the consistency of theta-band modulation in the human amygdala (Tang et al., 2021b) and hippocampus (Oehrn et al., 2015) suggest the possibility of an interaction between these areas during non-emotional conflict as well. Further studies on the communication between the amygdala and hippocampus during the Stroop task may be warranted to further understand their relationship in non-emotional conflict processing.

6.2.2. Cortico-limbic cooperation

It is believed that theta activity in the frontal cortex plays a large role in cognitive control and conflict processing (Cavanagh and Frank, 2014). Due to the observed activity of hippocampal (Oehrn et al., 2015) and amygdaloid (Tang et al., 2021b) power in the theta band during the Stroop task, it is plausible that limbic theta activity has a functional relationship with frontal cortical theta-band oscillations. Furthermore, prior MEG (Backus et al., 2016) and EEG (Babiloni et al., 2009) studies have provided evidence of theta coherence alterations between the hippocampus and cortical regions during cognitive processes, including memory recall and integration.

Our group observed decreases in theta coherence between the hippocampus and OFC during the post-stimulus (i.e., cue-processing) phase of the incongruent task condition (Tang et al., 2021a). Importantly, we observed no significant theta coherence changes in the congruent task condition or in the failed incongruent task trials (Tang et al., 2021a). This observed decrease in theta coherence is likely specific to successful conflict processing. Additionally, our group analysis showed that the difference in mean theta coherence values was larger when signal segments were aligned to the cue compared with those calculated from response-aligned segments (Tang et al., 2021a). This further characterizes the decreased theta coherence as having a more cognitive role than a motor one.

The functional significance of decreases in theta coherence during cognitive tasks like conflict processing remains an area of inquiry. Young et al. (2011) found a theta coherence decline between the OFC and hippocampus in rats presented with a discordant scenario (Young and Shapiro, 2011a). Thus, a decline in theta coherence may reflect a desynchronization for the brain regions to encode new task representations.

6.2.3. Orbitofrontal cortex

Oehrn et al. (2014) found that PAC between the theta and gamma bands within the DLPFC is indicative of successful conflict processing (Oehrn et al., 2014). We demonstrated that the OFC is also involved in conflict resolution via increased PAC between theta phase and gamma amplitude (Chen et al., 2022). This increase in PAC was strongest during the cue-processing period (after cue presentation and before verbal response), indicating the signal is cognitive. This work adds to the generally accepted theory that oscillations in the low-gamma range are local signals that can be modified by lower frequency bands such as the theta range (Buzsaki and Wang, 2012). Taken together with our separate study detailing theta decoherence between the hippocampus and OFC to resolve conflict (Tang et al., 2021a), it is clear that the neurophysiological interactions between the OFC and other regions involved in conflict (e.g., ACC and DLPFC) warrant further investigation to understand human conflict resolution.

Studies have found evidence of a relationship between the OFC and ACC in decision-making. Using fMRI, Cohen et al. (2005) found increased activity in the ACC and OFC of participants making a high-risk decision instead of a low-risk decision (Cohen, Heller and Ranganath, 2005). Using connectivity analyses, the authors also found significant functional connectivity between the OFC and ACC. Theta/low-beta synchronization has been observed between the OFC and ACC in rats during value-based decision making (Fatahi et al., 2018). Because theta-band power is thought to act over large regions of the brain (Oehrn et al., 2014), future studies should measure theta coherence between the OFC and ACC or DLPFC in the presence of conflict to determine how these structures interact to resolve conflict.

6.3. Direct neural recordings define a network in conflict processing

We propose a spatiotemporal network model of conflict processing that synthesizes the information obtained from intracranial recordings from other groups and our own group. First, in a condition involving conflict after cue presentation, the dACC serves as the detector of

conflict. At the single neuron level, there are small populations of neurons that encode task-specific and domain-general information based on their spike firing patterns before cue presentation, in the pre-response period, and after response (Sheth et al., 2012; Bonini et al., 2014; Smith et al., 2019; Ebitz et al., 2020; Fu et al., 2022a). At the population level (combining single units into pseudopopulations), the summed activity contains task-specific information that can differentiate between different types and levels of conflict (Ebitz et al., 2020; Fu et al., 2022a). These populations of neurons, as represented in the LFP, have increased gamma-band activity in response to conflict (Tang et al., 2016; Bartoli et al., 2018) and intra-regional gamma-band increases in coherence (Bartoli et al., 2018). There is some inconsistency as to whether theta-band activity is increased (Bartoli et al., 2018) or decreased (Tang et al., 2016) in the dACC during conflict processing, but most of the literature suggests an increase in theta-band power. Beta-band power was found to decrease in response to conflict (Tang et al., 2016; Bartoli et al., 2018). In response to conflict, the dACC recruits the DLPFC for conflict processing/resolution, as evidenced by observed gamma-theta cross frequency coupling (Bartoli et al., 2018) and spike-field coherence (SFC) between dACC neurons and DLPFC LFPs (Smith et al., 2019).

The DMPFC has also been implicated as an error-detecting brain region (Oehrn et al., 2014), but the exact temporal relationship and interaction with the dACC cannot be determined from the literature as no studies have evaluated the interaction between the dACC and DMPFC specifically. In response to conflict, theta and gamma power increase in the DLPFC. Conflict is associated with an increase in theta-gamma PAC between the DMPFC and DLPFC. In addition, there is information transfer in the theta range from the DMPFC to the DLPFC that occurs earlier in trials, while information is conveyed in the reverse direction later in trials, indicating a loop of conflict signaling.

Following activation of the dACC and DMPFC, the DLPFC is activated and appears to be an important structure in conflict processing and resolution. Here again, a small proportion of single neurons encode conflict by varying firing rates in response to conflict conditions, and this proportion is less than that in the dACC (Smith et al., 2019; Ebitz et al., 2020). The majority of DLPFC neurons show theta-range SFC, which was not seen in the dACC, indicating a different mode of conflict processing between the two structures; additionally, as mentioned earlier, there is SFC between the dACC and DLPFC (Smith et al., 2019). The population activity as determined by pseudopopulations of these neurons shows that the summed activity encodes task-specific information and can differentiate between different types and levels of conflict (Smith et al., 2019). Population activity as determined by LFPs shows that there is increased gamma-band activity in the DLPFC in response to conflict (Koga et al., 2011; Oehrn et al., 2014). Beta- and theta-band power decrease in the DLPFC in response to conflict (Tang et al., 2016; Bartoli et al., 2018). There is also theta-gamma PAC within the DLPFC in response to conflict signaling (Oehrn et al., 2014).

After activation of the DLPFC, other structures show activity as well. Single unit recordings from the pre-SMA show a similar relationship to the dACC in terms of conflict-responsive neurons, but they process conflict information after the dACC in the pre-response period and before the dACC in the post-response period (Fu et al., 2022a). The PMC (Koga et al., 2011; Caruana et al., 2014; Bartoli et al., 2018), SMA (Koga et al., 2011; Caruana et al., 2014), MFC (Tang et al., 2016), and OFC (Tang et al., 2016) showed increased gamma-band power in response to conflict, while there is a decrease in theta-power associated with conflict in the MFC (Tang et al., 2016) and OFC (Tang et al., 2016). The OFC shows intra-regional theta-gamma PAC (Tang et al., 2021a) as well as inter-regional theta coherence (Chen et al., 2022) in response to conflict.

The temporal dynamics of activation of the hippocampus, amygdala, and STN cannot be compared directly to those of the other frontal structures, as there have not been any studies involving recordings from these structures in the same experiments as other frontal structures, except for the OFC (Chen et al., 2022). In response to conflict, there is an increase in pre-response hippocampal low-gamma band power, a

pre-response increase followed by a post-response decrease in theta power in response to conflict, and a decrease in hippocampal theta-band power in a stimulus-locked analysis (Oehrn et al., 2015). There is also an increase in beta-band power in the hippocampus in the pre-response period (Chen et al., 2020). As mentioned earlier, there is theta coherence between the OFC and hippocampus (Chen et al., 2022). The amygdala also shows increased theta-band activity in the pre-response period in response to conflict (Tang et al., 2021b), and the STN also shows this activity in response to conflict (Brittain et al., 2012; Ghahremani et al., 2018).

The preceding paragraphs in this section refer mostly to the dynamics of a single trial, but there are also important effects that occur across subsequent conflict trials. Perhaps the most important phenomenon in this regard is the Gratton Effect. At the single neuron level, firing rates in dACC neurons were higher in trials preceded by conflict (Sheth et al., 2012). In terms of LFP analysis, the Gratton effect was observed in the gamma-band in the dACC, DLPFC, MFC, and OFC (Tang et al., 2016; Bartoli et al., 2018). There is also neuronal firing before cue presentation that can influence reaction time, and this information is domain-general in the dACC and task-specific in the DLPFC (Herman et al., 2023).

In terms of response to error, the pre-SMA and SMA appear to be the most important detector of error response, followed temporally by the dACC and then other structures in the MFC (Bonini et al., 2014; Fu et al., 2019). At the single neuron level, error is signaled through either increased spiking or decreased spiking, and some neurons signal if there was error in a previous trial (Fu et al., 2019). At the population level, as determined by analyzing pseudopopulation activity of single units, LFP recordings show increased theta-band power in the dACC and pre-SMA and decreased theta-band power in the hippocampus (Fu et al., 2019). There is also increased gamma-band power in the dACC, DLPFC, MFC, and OFC following self-corrected error trials (Tang et al., 2016).

Future studies will be needed to show how these brain regions interact within the network through studying coherence, cross-frequency coupling, and spike field coherence. The single unit recordings from the dACC, pre-SMA, and DLPFC have been helpful in defining conflict processing at the level of individual neurons, but in order to understand the whole network of conflict processing, single-unit recordings will need to be obtained from other frontal structures, the OFC, the hippocampus, and the amygdala.

7. Conclusion

We have summarized the studies on human conflict processing in order to define a network of frontal cortical structures, and our own studies also identified several novel neural oscillations associated with human conflict processing: response-locked beta increases in the hippocampus (Chen et al., 2020), response-locked theta increases in the amygdala (Tang et al., 2021b), decreased theta coherence between the OFC and hippocampus (Tang et al., 2021a), and increased theta-LG PAC in the OFC (Chen et al., 2022). Additional studies of direct neural recording in humans during conflict are needed to further map the interactions between brain regions involved in conflict processing and resolution. Studies comparing subjects with and without certain neuropsychiatric disorders will increase the clinical feasibility of using such markers to improve patient treatment.

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All authors meeting authorship criteria are listed, and all authors certify that sufficient participation was given in order to take public responsibility for the content contained within this manuscript. Furthermore, all authors certify that this material has not been submitted to or published any other publication. If accepted, this article will not be published elsewhere.

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Declaration of Competing Interest

None of the authors have potential conflicts of interest to be disclosed.

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